Radiative Transfer

Course A of two · Six concept-based, illustrated lectures for pre-master students, with worked examples and exam questions & answers

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Final Examination — Course A: Radiative Transfer (with complete solutions)
Part A — Short answers (2 marks each)
Part B — Problems
Part C — Formation of the RTE, polarization, history (4 marks each)
Part E — Essay (10 marks)

Formula Sheet — Course A: Radiative Transfer
Radiation field (Lecture RT1)
Black body (Lecture RT2)
Coefficients (Lecture RT2)

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Moments, diffusion, scattering (Lecture RT4)
Forming the RTE (Lecture RT3)
Historical approximations (Lecture RT6)
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Lectures RT1–RT6, final exam (RT7) and formula sheet (RT8). Every lecture opens with a concept map, explains each idea in words before the mathematics, and ends with misconceptions, a summary and four exam questions with answers. Course B, *Stellar Atmospheres*, applies this theory to stars.

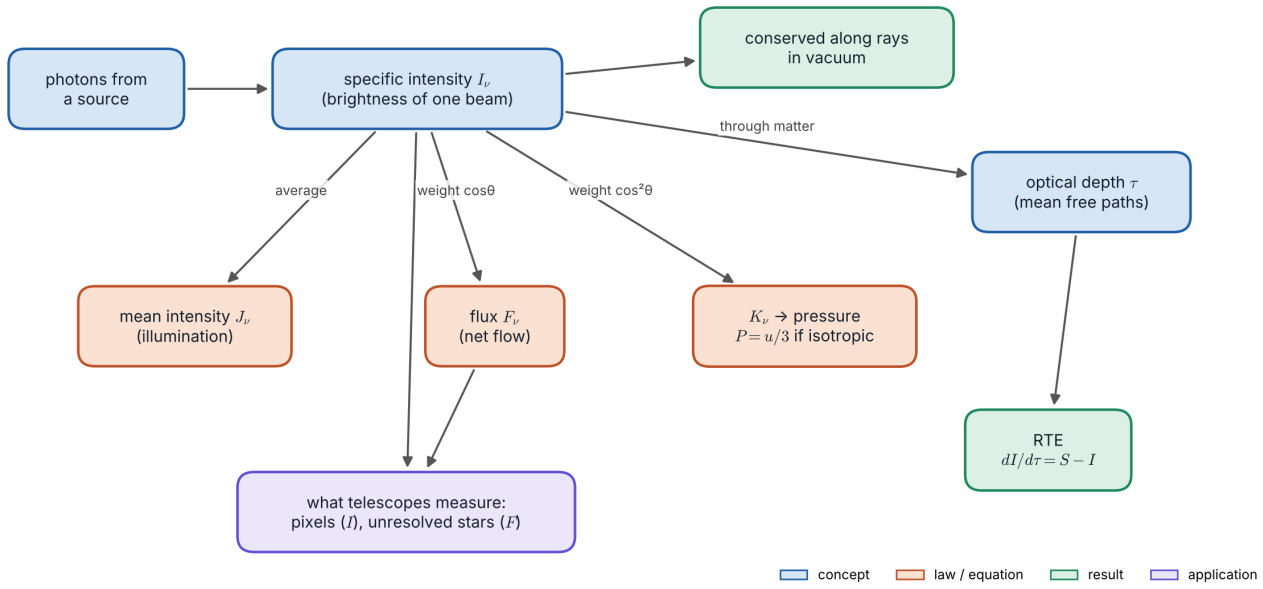
Based on: A670 *Advanced Astrophysics* Lecture 1 · W. Hu, *Astro 305 Notes, Set 1* · MIT 8.901 (2019) Lecture 7 · G. W. Collins II, *Fundamentals of Stellar Astrophysics*, Ch. 9 · S. Chandrasekhar, *Radiative Transfer* (1950) · E. Novotny, *Introduction to Stellar Atmospheres and Interiors* (1973)

Code: all figures and numbers are reproducible with the Python scripts in `scripts/`. The interactive web app is in `app/` and the spreadsheet is `Radiative_Transfer_Lab.xlsx`.

Lecture RT1 — Radiation and How We Describe It

> **Sources:** A670 *Advanced Astrophysics* Lecture 1; W. Hu, *Astro 305 Set 1*; MIT 8.901 (2019) Lecture 7; G. W. Collins II, *The Fundamentals of Stellar Astrophysics*, Ch. 9.2–9.3; S. Chandrasekhar, *Radiative Transfer* (1950), Ch. I; E. Novotny, *Introduction to Stellar Atmospheres and Interiors* (1973), Ch. 2.1.

RT1 concept map — describing radiation



*Concept map — read it from the top left. Everything starts with the **specific intensity**, the brightness of one beam. Averaging it over directions gives the mean intensity, the flux and the pressure. Sending it through matter introduces optical depth and leads to the transfer equation.*

Learning goals

1. Say in words what radiative transfer is and why astronomy depends on it.
2. Tell intensity apart from flux, and explain why intensity does not fall with distance.
3. Build J_ν, F_ν and K_ν from I_ν and give the physical meaning of each.
4. Read the transfer equation $dI_\nu/d\tau_\nu = S_\nu - I_\nu$ term by term.

The five big ideas of this lecture

1. **Radiative transfer is photon bookkeeping.** Follow one beam of light. Along the way matter *removes* photons (absorption, scattering out of the beam) and *adds* photons (emission, scattering into the beam). The transfer equation is this account: *change of the beam = gains − losses*.
2. **Intensity is the brightness of one beam.** It is what one camera pixel measures when it looks at an extended object.
3. **Intensity is conserved in empty space.** A lamp twice as far away delivers four times less energy, but it also looks four times smaller in area, so each patch of it looks equally bright.

4. **Flux is net flow.** It adds up the intensity from all directions, weighted by how squarely each beam crosses the surface. Light that is the same in all directions (isotropic) carries no net flux: a star shines only because its light is slightly *anisotropic*.
5. **Optical depth is distance measured in "fog units".** On a foggy day with 50 m visibility, an object 50 m away is at $\tau \approx 1$; at 500 m, $\tau \approx 10$, and it is gone. What we can see is set by τ, not by kilometres.

1.1 What radiative transfer does

Almost all we know about the Universe beyond the Solar System arrives as **photons** that have been emitted, absorbed and scattered by atoms, ions, electrons, molecules and dust. Radiative transfer (RT) connects the **physical state** of the gas (T, ρ, composition, velocity) with the **spectrum** we record.

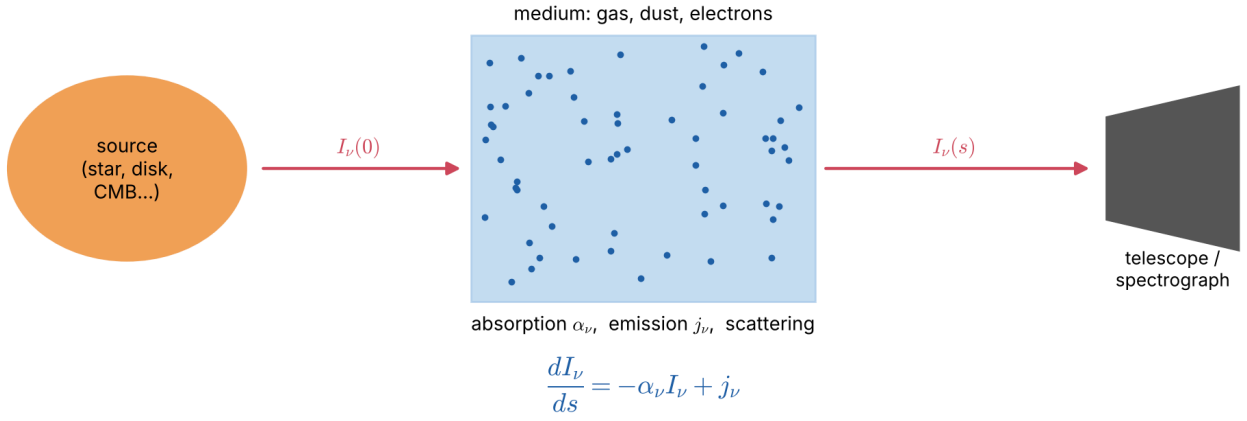
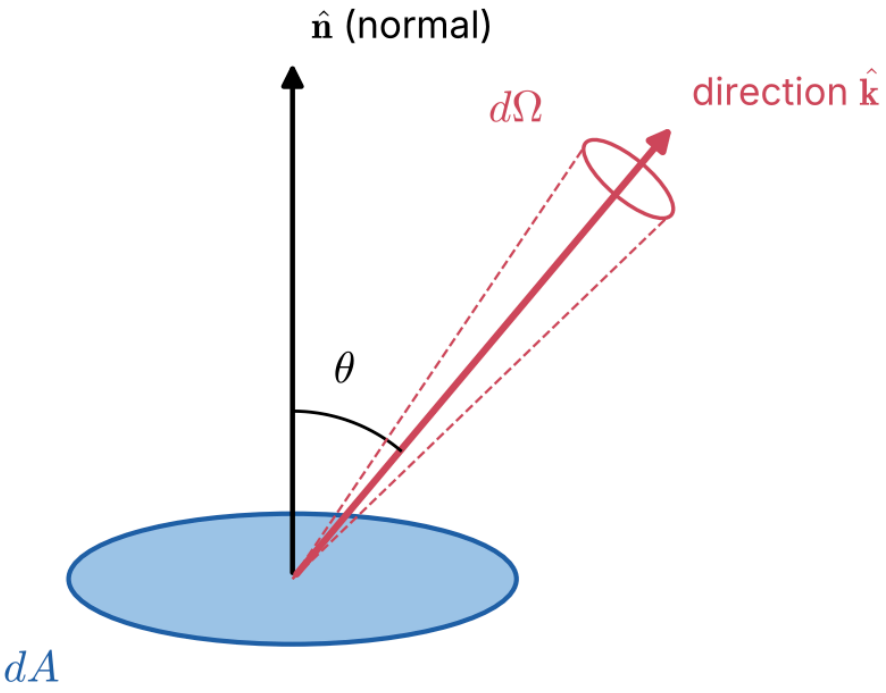


Figure 1.1 — A beam enters a medium. Absorption (α_ν) removes energy, emission (j_ν) adds energy, and scattering does both by redirecting photons. We observe what comes out.

The same theory is used in two directions:

Problem	Given	Find	Example
Forward (modelling)	$T(z)$, $\rho(z)$, composition	the emergent spectrum	a synthetic stellar spectrum
Inverse (diagnostics)	the observed spectrum	T, ρ, abundances	an abundance analysis

1.2 Specific intensity: the brightness of one beam



$$dE = I_\nu \cos\theta \, dA \, d\Omega \, d\nu \, dt$$

Figure 1.2 — Energy dE crosses an area dA, tilted by θ from the beam direction, into a small cone $d\Omega$, in frequency interval $d\nu$ and time dt.

$$\boxed{dE = I_\nu \ \cos\theta \ dA \ d\Omega \ d\nu \ dt} \qquad [I_\nu] = \mathrm{W\ m^{-2}\,Hz^{-1}\,sr^{-1}}.$$

In words: I_ν is energy per second, per square metre *facing the beam* (hence $\cos\theta$), per hertz, per steradian of sky. The **steradian** is the two-dimensional angle: the whole sky is 4π sr, and the Sun seen from Earth covers 6.8×10^{-5} sr.

Per unit wavelength, $I_\lambda = (c/\lambda^2) I_\nu$, and $\nu I_\nu = \lambda I_\lambda$. A spectrum's peak therefore sits at different places in the two representations.

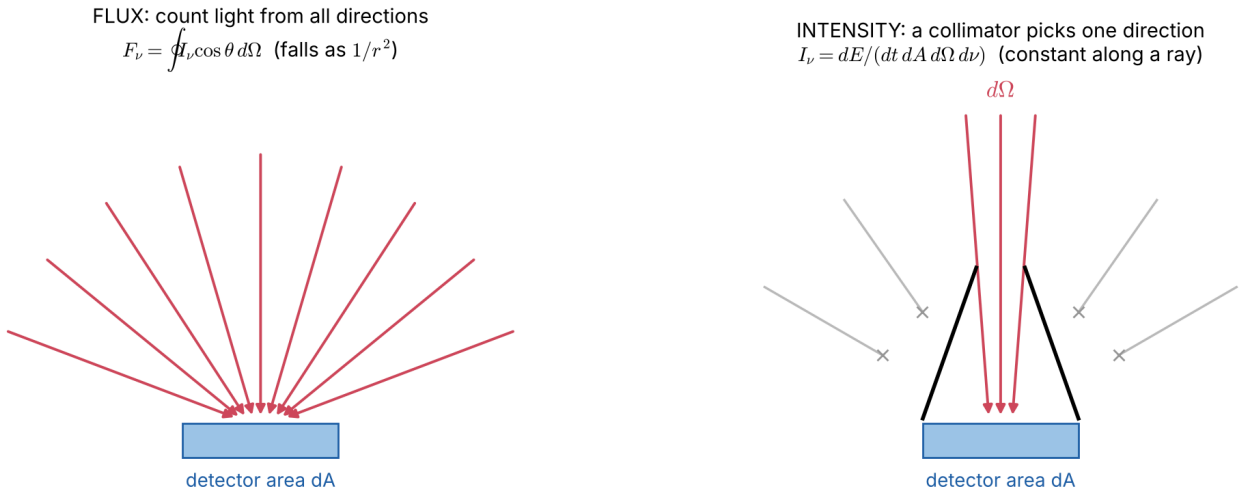


*Figure 1.3 — Left: a bare detector collects light from every direction, each beam weighted by $\cos\theta$; that is the **flux**. Right: a collimator (or one telescope pixel) admits only a narrow cone; that is the **intensity**.*

1.3 Why intensity does not fall with distance

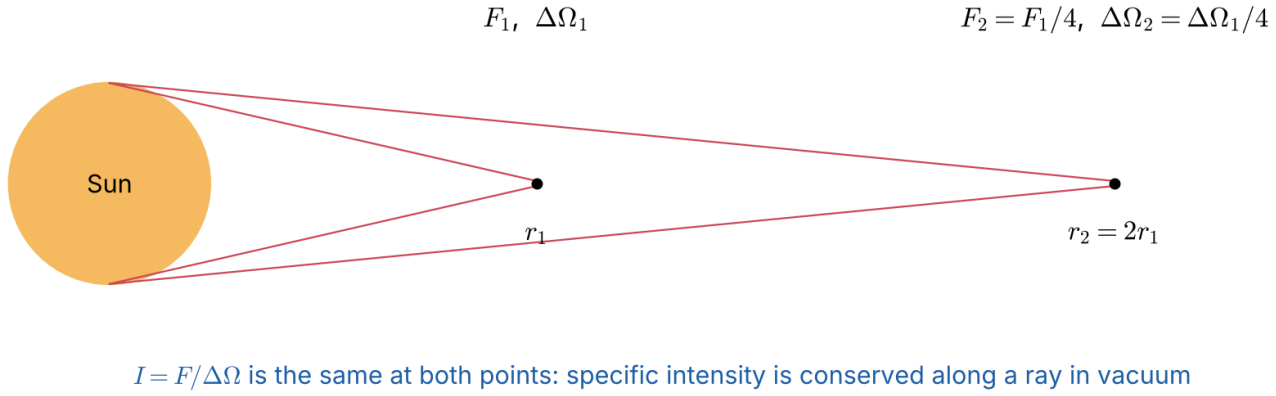


Figure 1.4 — The flux from the Sun falls as r^{-2}, and so does the solid angle it covers. Their ratio, the intensity, stays fixed.

Proof in two lines. Take two small areas dA_1, dA_2 a distance r apart, facing each other. The light crossing both leaves dA_1 into $d\Omega_1 = dA_2/r^2$ and arrives at dA_2 from $d\Omega_2 = dA_1/r^2$. With no matter in between, the energies are equal:

$$I_1\, dA_1 \frac{dA_2}{r^2} = I_2\, dA_2 \frac{dA_1}{r^2} \;\Rightarrow\; I_1 = I_2, \qquad \frac{dI_\nu}{ds} = 0 \text{ in vacuum.}$$

Consequence: a resolved object's surface brightness does not depend on its distance. (In an expanding Universe, I_ν/ν^3 is the conserved quantity, which is why surface brightness dims as $(1+z)^{-4}$.)

1.4 Moments: replacing directions by averages

Following I_ν in every direction is hard, so we use direction averages. With $\mu = \cos\theta$ and a plane-parallel field (no dependence on azimuth):

Moment	Definition	Plane-parallel form	Physical meaning
J_ν (0th)	$\frac{1}{4\pi}\oint I_\nu\, d\Omega$	$\frac{1}{2}\int_{-1}^{1} I_\nu\, d\mu$	how strongly a point is bathed in light
F_ν (1st)	$\oint I_\nu \cos\theta\, d\Omega$	$2\pi\int_{-1}^{1} I_\nu \mu\, d\mu$	net energy flow
H_ν (1st)	$F_\nu/4\pi$	$\frac{1}{2}\int_{-1}^{1} I_\nu \mu\, d\mu$	the “Eddington flux”
K_ν (2nd)	$\frac{1}{4\pi}\oint I_\nu \cos^2\theta\, d\Omega$	$\frac{1}{2}\int_{-1}^{1} I_\nu \mu^2\, d\mu$	momentum flow: pressure $P_\nu = 4\pi K_\nu/c$

The energy density is $u_\nu = 4\pi J_\nu/c$.

> **Three different “fluxes” in the literature** (Collins §9.3b): the physical flux F_ν, the Eddington flux $H_\nu = F_\nu/4\pi$ and the “astrophysical flux” $\mathcal{F}_\nu = F_\nu/\pi$. In this course F is always the physical flux, and $F = \sigma T_{\rm eff}^4$.

Three special fields to remember:

- **Isotropic** (I the same in all directions): $J = I$, $F = 0$, $K = J/3$, so $P = u/3$. Light is intense but there is no net flow, as deep inside a star.
- **One-sided** (uniform I outward, nothing inward, as at a stellar surface): $J = I/2$, $F = \pi I$, $K = I/6$. This is where the factor π in $F = \pi B = \sigma T^4$ comes from: slanted beams cross the surface less efficiently.
- **Linear in μ** ($I = I_0 + I_1\mu$): $J = I_0$, $H = I_1/3$, $K = I_0/3$. The field carries a flux, yet $K = J/3$ still holds exactly. This is the basis of the Eddington approximation (Lecture RT4).

Flux from a star. A sphere of radius R with uniform surface intensity I, seen from distance d, gives $F = \pi I (R/d)^2$. At its surface $F = \pi I$, far away $F \propto d^{-2}$, while I itself never changes.

1.5 Into matter: optical depth and the transfer equation (preview)

Inside matter, a beam loses a fixed *fraction* of itself per unit length (absorption) and gains energy from the gas whether or not light is present (emission):

$$\boxed{\frac{dI_\nu}{ds} = -\alpha_\nu I_\nu + j_\nu}$$

Symbol	Name	Units
$\alpha_\nu = n\sigma_\nu = \rho\kappa_\nu$	absorption coefficient	m^{-1}
κ_ν	opacity (per unit mass)	$\mathrm{m}^2\ \mathrm{kg}^{-1}$
j_ν	emission coefficient	$\mathrm{W\ m^{-3}\ Hz^{-1}\ sr^{-1}}$

Measuring distance in mean free paths, $d\tau_\nu = \alpha_\nu\, ds$, and defining the **source function** $S_\nu = j_\nu/\alpha_\nu$ gives the form used everywhere:

$$\frac{dI_\nu}{d\tau_\nu} = S_\nu - I_\nu.$$

Read it as a law of relaxation: if the beam is brighter than S_ν it fades; if fainter, it brightens. After a few optical depths the beam "forgets" where it came from and takes the value of S_ν. Lecture RT2 says what S_ν is made of; Lecture RT3 solves the equation.

TE, LTE and NLTE in one sentence each. In **TE** (a closed oven) matter and light share one temperature, and $I_\nu = B_\nu$. In **LTE**, collisions keep the *matter* at the local temperature, so $S_\nu = B_\nu(T)$, while the *light* streams freely and $I_\nu \neq B_\nu$. In **NLTE**, collisions are too rare and the radiation field itself sets the atomic populations. Course B (Lecture SA1) develops this with examples.

Where RT is used: stellar atmospheres (spectra, $T_{\rm eff}$, abundances), the ISM (extinction $A_V = 1.086\,\tau_V$, the 21-cm line), H II regions and planetary nebulae, planetary and exoplanet atmospheres, accretion disks and AGN, the CMB, and stellar interiors (the diffusion limit and the Rosseland mean).

Worked example 1.1 — The Sun: from luminosity to intensity

Given $L_\odot = 3.828 \times 10^{26}$ W, $R_\odot = 6.957 \times 10^{8}$ m and 1 AU $= 1.496 \times 10^{11}$ m, find the surface flux, $T_{\rm eff}$, the mean intensity of the disk, and the solar constant computed two ways.

Solution.

- $F_\odot = L_\odot / 4\pi R_\odot^2 = 6.294 \times 10^{7}$ W m^{-2}, so $T_{\rm eff} = (F/\sigma)^{1/4} = 5772$ K.
- $\bar{I} = F/\pi = 2.003 \times 10^{7}$ W m^{-2} sr^{-1}.
- The Sun's solid angle at 1 AU is $\Omega_\odot = \pi (R_\odot/\mathrm{AU})^2 = 6.794 \times 10^{-5}$ sr. Intensity is conserved, so $F_\oplus = \bar{I}\,\Omega_\odot = 1361$ W m^{-2}. The inverse-square law gives the same: $F_\odot (R_\odot/\mathrm{AU})^2 = 1361$ W m^{-2}.

Worked example 1.2 — Extinction through a slab

A beam crosses a slab of thickness $L = 10^{12}$ m with $\alpha_\nu = 2 \times 10^{-12}$ m^{-1} and no emission. What fraction emerges, and what is the extinction in magnitudes?

Solution. $\tau_\nu = \alpha_\nu L = 2$, so the transmitted fraction is $e^{-2} = 0.135$. In magnitudes, $A = 2.5 \log_{10} e^{\tau} = 1.086\,\tau = 2.17$ mag.

Worked example 1.3 — Radiation pressure at the Sun's centre

At the centre, $T_c = 1.57 \times 10^{7}$ K and $P_{\rm gas} \approx 2.3 \times 10^{16}$ Pa. Find $P_{\rm rad}$.

Solution. The radiation there is in TE and isotropic, so $P_{\rm rad} = u/3 = aT^4/3$ with $a = 4\sigma/c = 7.566 \times 10^{-16}$ J m^{-3} K^{-4}. This gives $P_{\rm rad} = 1.5 \times 10^{13}$ Pa, only 7×10^{-4} of the gas pressure. Radiation pressure matters only in very massive stars.

Common misconceptions

1. **"Intensity falls as $1/r^2$."** Flux does; intensity (surface brightness) does not, in empty space.

2. **“Zero flux means no light.”** An isotropic field can be extremely intense and still carry no net flux.
3. **“Optical depth is a distance.”** It counts mean free paths. One kilometre of air and one millimetre of stellar interior can have wildly different τ.
4. **“In LTE the intensity equals the Planck function.”** In LTE the *source function* equals $B_\nu(T)$; the intensity follows from the transfer equation and generally differs from B_ν.

Lecture summary

- RT is bookkeeping of photons gained and lost along a beam.
- I_ν is the brightness of one beam, and it is conserved along rays in vacuum.
- J_ν, F_ν and K_ν are the 0th, 1st and 2nd moments: illumination, net flow and pressure. Isotropic light has $F = 0$ and $P = u/3$; a uniformly bright surface emits $F = \pi I$.
- In matter, $dI_\nu/d\tau_\nu = S_\nu - I_\nu$: the beam relaxes toward the source function over about one optical depth.

Exam questions and answers — Lecture RT1

Q1. Prove that I_ν is constant along a ray in vacuum. Hence explain why the Sun’s disk looks equally bright from Earth and from Mars.

> **A1.** Energy conservation between two areas dA_1, dA_2 a distance r apart gives $I_1 dA_1 (dA_2/r^2) = I_2 dA_2 (dA_1/r^2)$, so $I_1 = I_2$. From Mars (1.52 AU) the flux is lower by $1/1.52^2 = 0.43$, but the Sun’s solid angle is smaller by the same factor. The brightness per steradian, I_ν, is unchanged.

Q2. A surface emits $I(\mu) = I_0$ for $\mu > 0$ and nothing for $\mu < 0$. Find J, F and K at the surface. Why is $F = \pi I_0$ and not $2\pi I_0$?

> **A2.** $J = \frac{1}{2}\int_0^1 I_0\, d\mu = I_0/2$; $F = 2\pi \int_0^1 I_0 \mu\, d\mu = \pi I_0$; $K = \frac{1}{2}\int_0^1 I_0 \mu^2\, d\mu = I_0/6$. The outward hemisphere is 2π sr, but each beam is weighted by $\cos\theta$, the projected area it crosses; the average of μ over the hemisphere is $1/2$. Note $J = 2H$, Eddington’s surface condition (Lecture SA4).

Q3. What is the difference between TE and LTE? Is the statement “in LTE, $I_\nu = B_\nu$” true?

> **A3.** In TE matter and radiation are both in equilibrium at one temperature: $I_\nu = J_\nu = B_\nu(T)$, isotropic, with no net flux. In LTE only the *matter* is in equilibrium at the local T (Maxwell, Boltzmann and Saha hold), so $j_\nu = \alpha_\nu B_\nu$ and $S_\nu = B_\nu(T)$. The statement is **false**: at a stellar surface, for instance, $I_\nu(\mu < 0) = 0$ while $B_\nu \neq 0$.

Q4. A medium has $\tau_\nu = 0.01$ at one frequency and $\tau_\nu = 100$ at another. Describe what an observer sees at each.

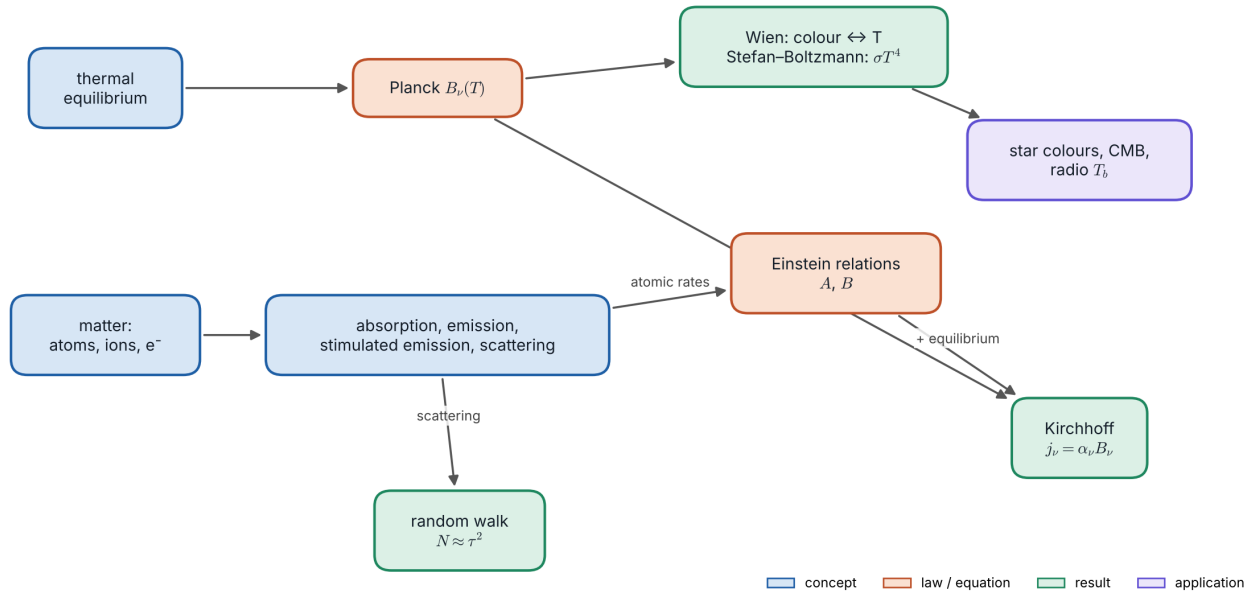
> **A4.** At $\tau = 0.01$ (optically thin) 99 % of the background passes, and the medium adds only $\approx \tau S_\nu$: we see through it, and its own glow is faint and proportional to its column density. At $\tau = 100$ (optically thick) the

> background is blocked (e^{-100}), and we see only the medium's outer layers, $I_\nu \approx S_\nu$ evaluated near $\tau \approx 1$.

Lecture RT2 — Black Bodies and How Light Interacts with Matter

> **Sources:** MIT 8.901 Lec. 7 (Stefan–Boltzmann, Wien, $\alpha_\nu = n\sigma_\nu = \rho\kappa_\nu$); Hu, *Astro 305 Set 1* (emission, absorption, scattering, random walk); Collins Ch. 9.1–9.2; Chandrasekhar Ch. I §§7–11; Novotny Ch. 2.2; Rybicki & Lightman Ch. 1.

RT2 concept map — black bodies and light–matter interaction



*Concept map — two threads meet in this lecture. The top thread is **light in equilibrium**: the Planck function and its laws. The bottom thread is **matter**: the processes by which atoms absorb, emit and scatter. Einstein's relations tie the two together, and Kirchhoff's law follows.*

Learning goals

1. Explain why the Planck spectrum depends only on temperature, and use Wien's and Stefan–Boltzmann's laws.
2. Name the four light–matter interactions and the coefficients α_ν, j_ν, σ_ν, κ_ν that describe them.
3. Derive the Einstein relations and the stimulated-emission correction.
4. Distinguish absorption from scattering, and write $S_\nu = \varepsilon B_\nu + (1-\varepsilon) J_\nu$.
5. Estimate photon escape times with the random walk.

The big ideas in plain words

A black body is a perfect absorber, and therefore a perfect emitter. The best example is a closed oven with a tiny hole. Light entering the hole is absorbed long before it can come out, so the hole looks black when the oven is cold. When the oven is hot, the light leaking out has a spectrum that depends **only on temperature**, not on what the oven is made of. That universal spectrum is the Planck function $B_\nu(T)$.

Why the Planck curve rises and falls. At higher frequency there are more "slots" (modes) for photons, growing as ν^2. But each photon costs $h\nu$, and the thermal energy kT cannot pay for expensive photons: their occupation falls as $e^{-h\nu/kT}$. Rising slots times falling occupation gives a peak.

Colour is a thermometer, brightness a steep one. The peak moves to shorter wavelength as T rises (Wien). Total power grows as T^4 (Stefan–Boltzmann): double the temperature and the power rises sixteen-fold.

Matter talks to light in four ways (Figure 2.2): a photon is absorbed by an atom; an excited atom emits spontaneously; a passing photon *stimulates* an excited atom to emit a twin; or a photon simply bounces off (scattering).

Sponge versus pinball. In *true absorption* the photon's energy becomes heat in the gas, and the gas later emits according to its temperature. In *scattering* the photon only changes direction, keeping its energy; it does not "know" the gas temperature. The number ε is the probability that an interaction is a true absorption.

Light leaves stars by a drunkard's walk. After N random steps a photon is only $\sqrt{N}$ steps from where it started, so escaping from optical depth τ takes about τ^2 steps.

2.1 The Planck function

Radiation inside an opaque cavity in thermodynamic equilibrium (*Hohlraum* radiation) is isotropic, unpolarised, and depends only on T:

$$\boxed{B_\nu(T) = \frac{2h\nu^3}{c^2}\,\frac{1}{e^{h\nu/kT}-1}} \qquad \boxed{B_\lambda(T) = \frac{2hc^2}{\lambda^5}\,\frac{1}{e^{hc/\lambda kT}-1}}$$

Derivation in three steps.

1. Count the modes: the number per unit volume in $d\nu$ is $8\pi\nu^2/c^3$, including two polarisations.
2. Find the occupation: photons are bosons with zero chemical potential, so $\bar{n} = 1/(e^{h\nu/kT}-1)$.
3. Combine: $u_\nu = (8\pi\nu^2/c^3)\,h\nu\,\bar{n}$. For isotropic light $u_\nu = 4\pi I_\nu/c$ (Lecture RT1), so $I_\nu = B_\nu$.

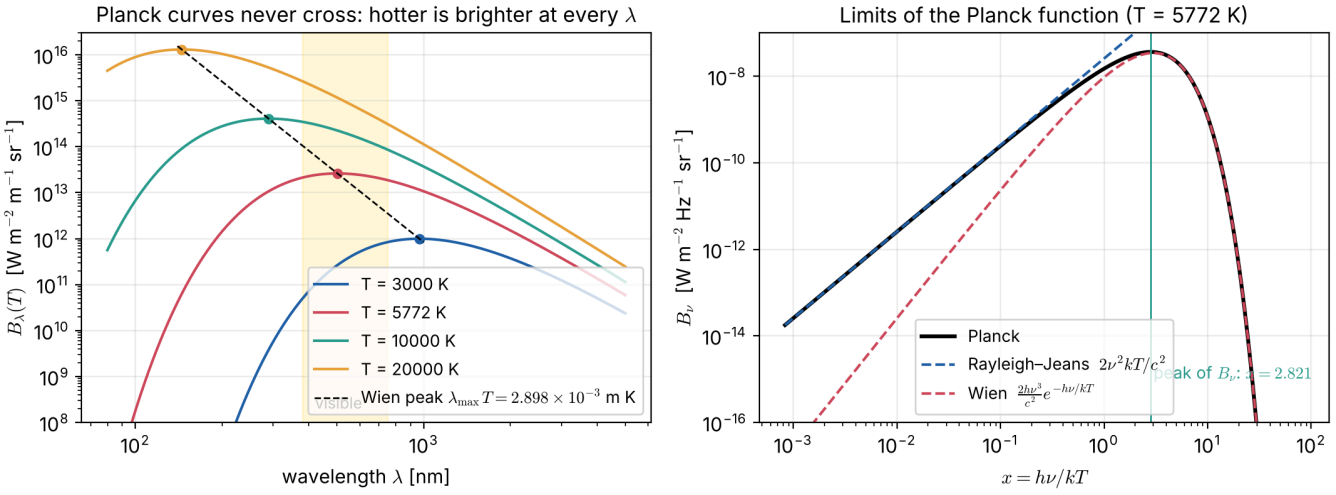


*Figure 2.1 — Left: B_λ at four temperatures, with the Wien peak marked. A hotter body is brighter at **every** wavelength, so Planck curves never cross. Right: the Rayleigh–Jeans and Wien limits.*

Law	Formula	Meaning
Rayleigh–Jeans ($h\nu \ll kT$, radio)	$B_\nu \approx 2\nu^2 kT/c^2$	linear in T, so radio astronomers quote **brightness temperatures**
Wien tail ($h\nu \gg kT$, UV)	$B_\nu \approx (2h\nu^3/c^2)e^{-h\nu/kT}$	exponentially sensitive to T
Wien displacement	$\lambda_{\max} T = 2.898 \times 10^{-3}$ m K; $h\nu_{\max} = 2.821\,kT$	colour ↔ temperature; note $\lambda_{\max} \neq c/\nu_{\max}$
Stefan–Boltzmann	$F = \pi B = \sigma T^4$, $\sigma = 2\pi^5 k^4/15c^2h^3$	total power
Energy density, pressure	$u = aT^4$, $P = u/3$, $a = 4\sigma/c$	radiation as a gas
Photon gas	$n_\gamma = 2.03 \times 10^7\, T^3$ m^{-3}, $\langle h\nu \rangle = 2.70\,kT$	

Many temperatures, one per question.

Temperature	Defined by	Question it answers
effective T_{eff}	$F = \sigma T_{\text{eff}}^4$	which black body emits the same total flux?
brightness $T_b(\nu)$	$I_\nu = B_\nu(T_b)$	which black body is equally bright at this ν?
colour T_c	the spectral slope	which black body has the same shape?
excitation T_{ex}	$n_2/n_1 = (g_2/g_1)e^{-h\nu/kT_{\text{ex}}}$	which T explains these level populations?

They coincide only for a true black body.

2.2 The four interactions

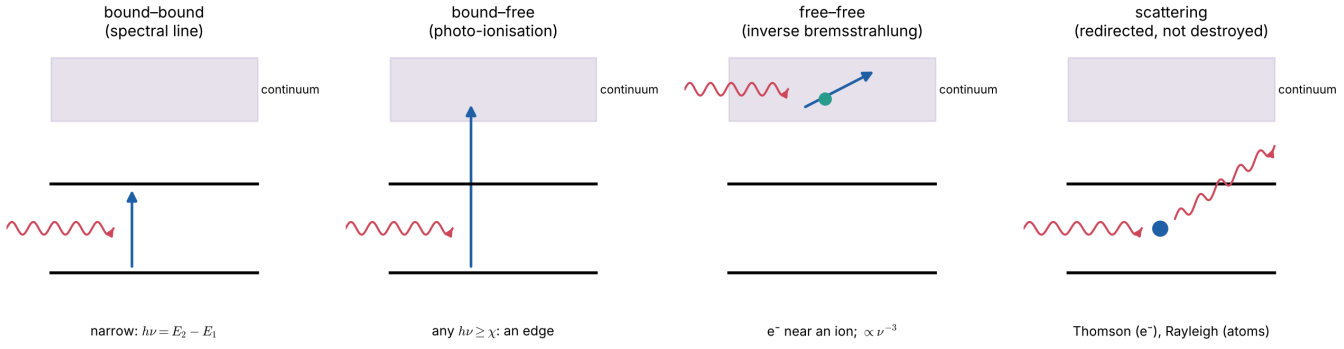


Figure 2.2 — Where opacity comes from. ***Bound–bound:*** *a narrow line at* $h\nu = E_u - E_l$. ***Bound–free:*** *any photon above the ionisation energy, which gives an absorption*
edge
. ***Free–free:*** *a free electron absorbs a photon while passing an ion.* ***Scattering:*** *the photon is redirected, not destroyed.*

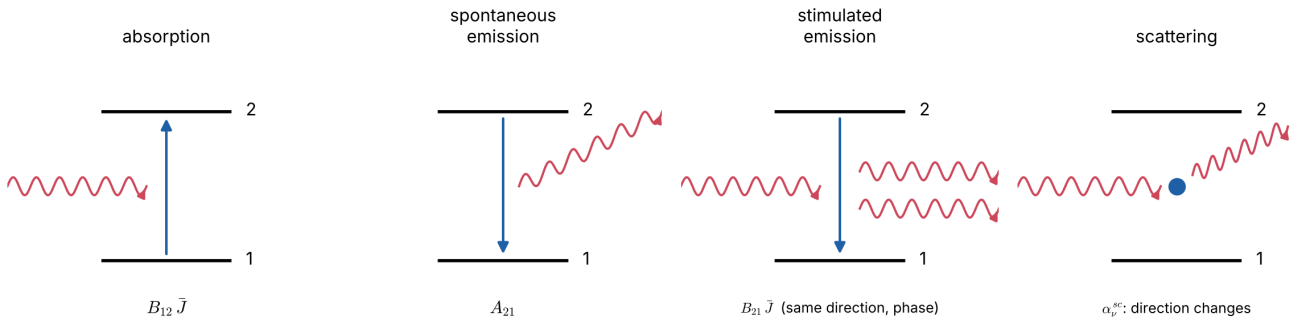


*Figure 2.3 — Absorption, spontaneous emission, stimulated emission and scattering in a two-level atom. Stimulated emission copies the incoming photon exactly, so it acts as **negative absorption**.*

Coefficients. The absorption coefficient is the probability per metre that a photon is removed:

$$\boxed{\alpha_\nu = n\sigma_\nu = \rho\kappa_\nu}, \qquad \ell_\nu = 1/\alpha_\nu.$$

Here σ_ν is the *effective target area* of one absorber (an electron has no size, yet $\sigma_T = 6.652 \times 10^{-29}$ m^2), and κ_ν is the opacity per kilogram. The emission coefficient j_ν is the power emitted per volume, frequency and steradian; for isotropic emission $j_\nu = P_\nu/4\pi$.

The opacity sources of stars (details in Lecture SA2):

Process	Frequency dependence	Where it matters
bound–bound (lines)	narrow profile $\phi(\nu)$	spectral lines
bound–free	edges, $\propto \nu^{-3}$ above each	Balmer jump; H$^-$ in the Sun
free–free	$\propto \nu^{-3}$ (optical), ν^{-2} (radio)	hot stars, H II regions
Thomson scattering	gray	O stars, interiors, disks; $\kappa_{\rm es} = 0.020(1+X)$ m^2 kg^{-1}

2.3 Einstein coefficients: linking matter to the Planck function

For a transition $1 \leftrightarrow 2$ with $h\nu_0 = E_2 - E_1$, weights g_1, g_2 and profile $\phi(\nu)$ (with $\int \phi\, d\nu = 1$), the rates per unit volume are: spontaneous emission $n_2 A_{21}$, absorption $n_1 B_{12} \bar{J}$ and stimulated emission $n_2 B_{21} \bar{J}$, where $\bar{J} = \int J_\nu \phi\, d\nu$.

The argument. In TE, detailed balance holds, the populations are Boltzmann, and $\bar{J} = B_\nu(T)$ at *every* temperature. That is possible only if

$$\boxed{g_1 B_{12} = g_2 B_{21}, \qquad A_{21} = \frac{2h\nu^3}{c^2} B_{21}}$$

These are properties of the atom, so they hold out of equilibrium too.

Macroscopic coefficients and the line source function:

$$j_\nu = \frac{h\nu}{4\pi} n_2 A_{21} \phi, \qquad \alpha_\nu = \frac{h\nu}{4\pi} n_1 B_{12} \left(1 - \frac{g_1 n_2}{g_2 n_1}\right) \phi, \qquad S_\nu = \frac{2h\nu^3}{c^2} \frac{1}{\dfrac{g_2 n_1}{g_1 n_2} - 1}.$$

- The bracket is the **stimulated-emission correction**. If $n_2/g_2 > n_1/g_1$ (an inversion), $\alpha_\nu < 0$ and light is amplified: a **maser**.
- With Boltzmann populations at T, $S_\nu = B_\nu(T)$ exactly. This is **Kirchhoff's law**, $j_\nu = \alpha_\nu B_\nu$, the foundation of LTE (Lecture SA1).

2.4 Scattering and the mixed source function

For coherent, isotropic scattering, energy taken from all directions is re-emitted evenly: $j_\nu^{\rm sc} = \alpha_\nu^{\rm sc} J_\nu$, so $S_\nu^{\rm sc} = J_\nu$. The transfer equation becomes an **integro-differential equation**, because J depends on I in all directions.

With both processes the coefficients add, and

$$\boxed{S_\nu = \varepsilon_\nu B_\nu + (1-\varepsilon_\nu) J_\nu, \qquad \varepsilon_\nu = \frac{\alpha_\nu^a}{\alpha_\nu^a + \alpha_\nu^{\rm sc}}}$$

where $1-\varepsilon$ is the single-scattering albedo. This one equation is the seed of all non-LTE physics (Lectures RT4 and SA3).

2.5 Random walk and thermalisation

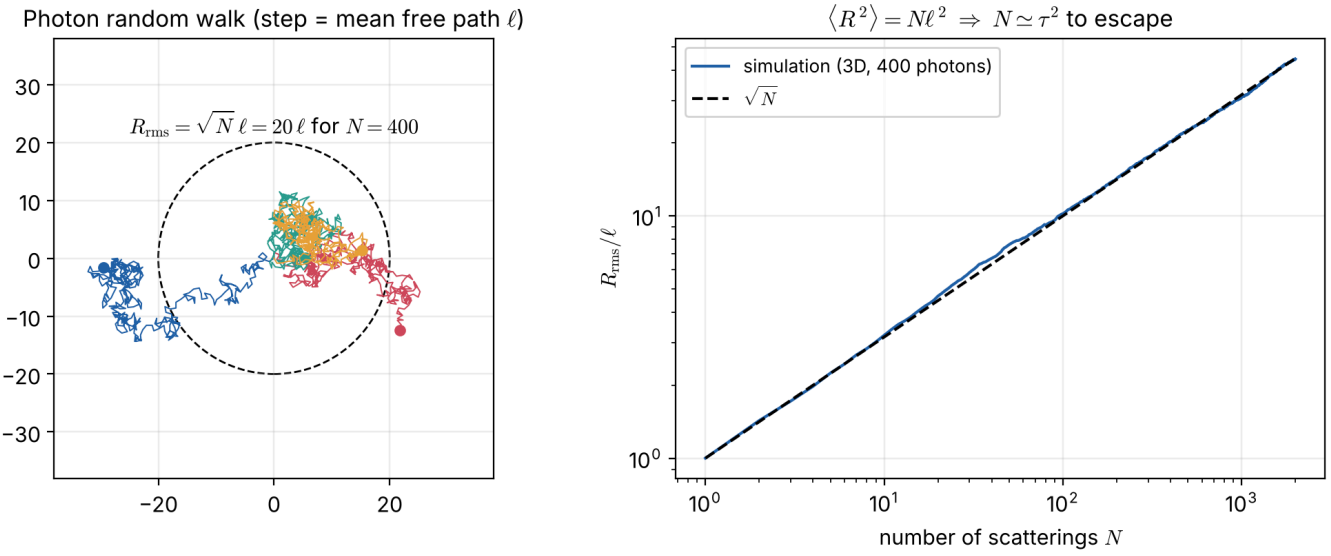


Figure 2.4 — Left: random photon paths; the dashed circle has radius $\sqrt{N}\ell$. Right: a 3D simulation confirms $R_{\rm rms} = \sqrt{N}\,\ell$.

Random unit steps give $\langle R^2 \rangle = N\ell^2$, so escaping from optical depth τ takes $N \approx \tau^2$ steps (or $\approx \tau$ if $\tau < 1$). If each interaction destroys the photon with probability ε, it survives about $1/\varepsilon$ steps and wanders a net optical distance $\tau_{\rm th} \approx 1/\sqrt{\varepsilon}$: the **thermalisation depth**, a key idea in Lecture RT4.

Worked example 2.1 — The cosmic microwave background

$T = 2.725$ K. *Find the peaks of B_ν and B_λ and the photon density.*

Solution.

- $\nu_{\max} = 5.879 \times 10^{10} \times 2.725 = 160$ GHz.
- $\lambda_{\max} = 2.898 \times 10^{-3}/2.725 = 1.06$ mm. Note that $c/\nu_{\max} = 1.87$ mm is a different number.
- $n_\gamma = 2.03 \times 10^7 \times 2.725^3 = 4.1 \times 10^8$ m^{-3}, which is **411 photons per cm³**.

Worked example 2.2 — Brightness temperature of a radio source

At 1.4 GHz a source has $I_\nu = 2.0 \times 10^{-19}$ W m^{-2} Hz^{-1} sr^{-1}. Find T_b.

Solution. $h\nu/k = 0.067$ K, so Rayleigh–Jeans applies and $T_b = c^2 I_\nu/(2k\nu^2) = 332$ K. If it is an optically thin H II region with $T_e = 8000$ K, then $T_b \approx T_e\tau$ (Lecture RT3), so $\tau \approx 0.04$.

Worked example 2.3 — How long does light take to leave the Sun?

Uniform Sun, fully ionised, $X = 0.7$, Thomson scattering only.

Solution.

1. $\bar\rho = M_\odot/(\tfrac{4}{3}\pi R_\odot^3) = 1410$ kg m^{-3} and $n_e = \bar\rho(1+X)/(2m_H) = 7.2 \times 10^{29}$ m^{-3}.
2. $\ell = 1/(n_e\sigma_T) = 2.1$ cm, so $\tau = R_\odot/\ell = 3.3 \times 10^{10}$.
3. $t = N\ell/c = R_\odot^2/(\ell c) = 7.7 \times 10^{10}$ s $\approx$ **2400 yr**, against 2.3 s for straight flight.

(Real opacities in the core are higher, giving 10^4–10^5 yr.)

Worked example 2.4 — When is stimulated emission important?

Compare the correction $1 - e^{-h\nu/kT}$ for the 21-cm line at 100 K and for Hα in the Sun.

Solution. For 21 cm, $h\nu/kT = 6.8 \times 10^{-4}$: stimulated emission cancels **99.93 %** of the absorption. For Hα at 5772 K, $h\nu/kT = 3.8$, and the factor is 0.98, only a 2 % correction. The rule: stimulated emission matters whenever $h\nu \lesssim kT$, i.e. in the radio.

Common misconceptions

1. **“Black bodies look black.”** Only when cold. The Sun is very nearly a black body.
2. **“Brightness temperature is the real temperature.”** Only for an optically thick thermal source. Thin gas has $T_b \approx \tau T < T$; masers can have T_b far above any physical temperature.
3. **“Scattering removes energy from the radiation.”** It takes photons from one beam and puts them in others; no energy is lost to the gas.
4. **“A photon from the core reaches us directly.”** Energy random-walks out over thousands of years, absorbed and re-emitted countless times.

Lecture summary

- $B_\nu(T)$ is the spectrum of light in equilibrium. Wien: $\lambda_{\max}T = 2.898$ mm K. Stefan–Boltzmann: $F = \sigma T^4$.

- Matter interacts through bound–bound, bound–free, free–free and scattering processes, described by $\alpha_\nu = n\sigma_\nu = \rho\kappa_\nu$ and j_ν.
- Einstein relations are atomic properties. With Boltzmann populations they give Kirchhoff's law, $S_\nu = B_\nu$.
- Stimulated emission is negative absorption; it matters in the radio and makes masers possible.
- Scattering gives $S = J$; mixing gives $S = \varepsilon B + (1-\varepsilon)J$; escape takes $\sim \tau^2$ steps.

Exam questions and answers — Lecture RT2

Q1. Derive the Planck function from the density of photon states and Bose–Einstein statistics, and show that its peak lies at $h\nu = 2.821\,kT$.

> **A1.** Modes per unit volume per $d\nu$: $8\pi\nu^2/c^3$. Mean energy per mode: $h\nu/(e^{h\nu/kT}-1)$. So $u_\nu = (8\pi h\nu^3/c^3)/(e^{h\nu/kT}-1)$ and $B_\nu = cu_\nu/4\pi$. With $x = h\nu/kT$, $B_\nu \propto x^3/(e^x-1)$; setting $d/dx = 0$ gives $3(1-e^{-x}) = x$, with root $x = 2.821$ (iterate $x_{n+1} = 3(1-e^{-x_n})$).

Q2. Derive the Einstein relations. Why do they hold even when the gas is not in TE?

> **A2.** In TE, $n_1 B_{12}\bar{J} = n_2 A_{21} + n_2 B_{21}\bar{J}$ with Boltzmann populations gives $\bar{J} = (A_{21}/B_{21})/[(g_1 B_{12}/g_2 B_{21})e^{h\nu/kT} - 1]$. This must equal $B_\nu(T)$ at every T, which requires $g_1 B_{12} = g_2 B_{21}$ and $A_{21} = (2h\nu^3/c^2)B_{21}$. The coefficients depend only on atomic wavefunctions, not on the environment, so they hold everywhere.

Q3. Derive $S_\nu = \varepsilon B_\nu + (1-\varepsilon)J_\nu$ and explain the meaning of ε.

> **A3.** $j = \alpha^a B + \alpha^{\mathrm{sc}} J$ (thermal emission by Kirchhoff, plus scattering) and $\alpha = \alpha^a + \alpha^{\mathrm{sc}}$. Dividing gives the result, with $\varepsilon = \alpha^a/(\alpha^a + \alpha^{\mathrm{sc}})$ the probability that an interaction destroys the photon (turns it into heat).

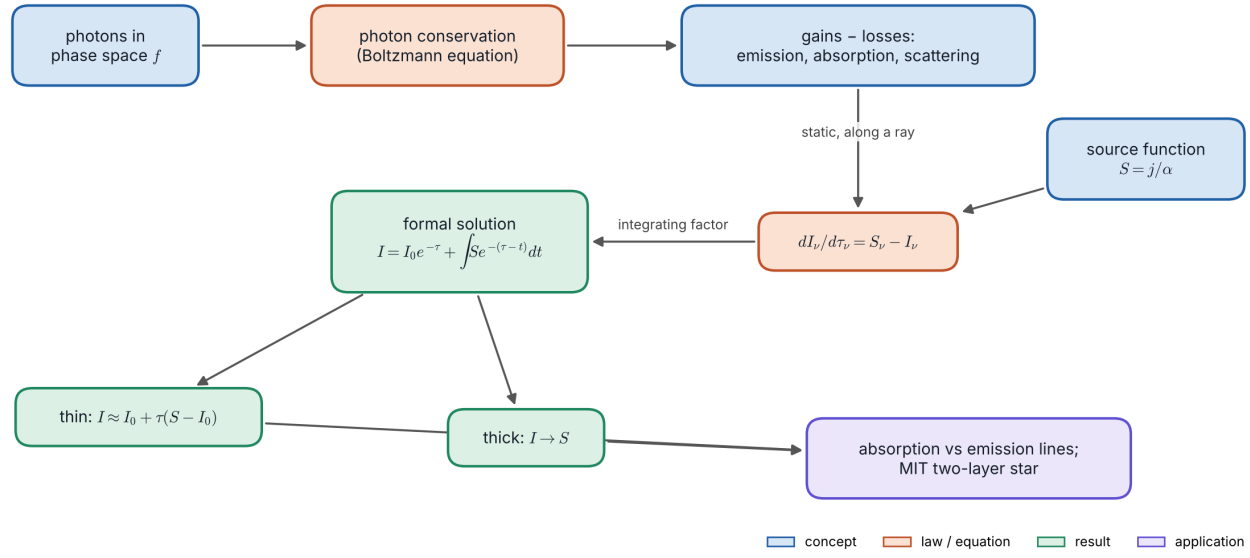
Q4. A cloud has $\tau = 100$ for pure scattering. About how many scatterings does a photon undergo before escaping, and how far does it travel in total?

> **A4.** $N \approx \tau^2 = 10^4$. The total path is $N\ell = \tau^2\ell = 100\,L$, where $L = \tau\ell$ is the cloud size, so the escape time is $100\,L/c$.

Lecture RT3 — The Transfer Equation: Where It Comes From and How to Solve It

> **Sources:** MIT 8.901 Lec. 7 §§7.1–7.3 (the RTE built in three steps; Kirchhoff's law for thick and thin lumps; thin H cloud; two-layer star; thermal inversions); Hu, *Astro 305 Set 1* (Liouville's theorem, formal solution); Collins Ch. 9.2 (Boltzmann equation → general RTE, redistribution function); Chandrasekhar Ch. I; Novotny Ch. 2.3–2.4; Rybicki & Lightman Ch. 1.

RT3 concept map — forming and solving the transfer equation



*Concept map — the top row **forms** the equation: photons are particles, their number is conserved, and matter adds and removes them. The middle row writes it along a ray. The bottom row **solves** it with an integrating factor and reads off the thin and thick limits, which explain why spectra show absorption or emission lines.*

Learning goals

1. Derive the transfer equation from photon conservation and name every gain and loss term.
2. Know which assumption leads to which special form (static, ray, plane-parallel, spherical).
3. Derive the formal solution and interpret it as "light from every layer, dimmed by the layers in front".
4. Solve the uniform slab, and use it to explain absorption lines, emission lines and thermal inversions.

The big ideas in plain words

The equation is just conservation. Photons are particles. In empty space they fly straight and nothing changes. Matter adds photons (emission, scattering *into* the beam) and removes them (absorption, scattering *out of* the beam). Rate of change along the path = gains − losses. That is the whole equation.

The equation is a tug of war. If the beam is brighter than the source function S_ν, absorption wins and the beam fades; if fainter, emission wins and it brightens. After about one optical depth the beam forgets its origin and approaches S_ν.

The solution is a foggy street of lamps. Looking down a foggy street, the light reaching you is the sum of every lamp, each dimmed by the fog between it and you. Near lamps dominate; far lamps fade as $e^{-\tau}$. So we "see" about one optical depth into anything (Figure 3.4).

Thin versus thick. A thin medium ($\tau \ll 1$) is a light mist: you see through it, and its own glow is proportional to how much gas there is. A thick medium ($\tau \gg 1$) is a wall: you see only its surface, glowing at S_ν.

Lines are thermometers of height. At line frequencies the gas is more opaque, so we see *higher* layers. If the temperature falls outward, the line looks darker than the continuum: an absorption line. If it rises outward (a chromosphere), the line is in emission.

3.1 Forming the equation

Step 1 — describe the photons

Count photons in phase space: $dN = f(\mathbf{r}, \mathbf{p}, t)\, d^3x\, d^3p$. With $p = h\nu/c$, the energy crossing dA in dt is $h\nu f\,(c\,dt\,dA\cos\theta)(h^3\nu^2/c^3)\,d\nu\,d\Omega$. Comparing with the definition of I_ν:

$$\boxed{I_\nu = \frac{h^4\nu^3}{c^2}\, f} \qquad \text{(Collins eq. 9.2.8)}$$

In equilibrium, $f = (2/h^3)/(e^{h\nu/kT} - 1)$, and this gives exactly B_ν, a useful check.

Step 2 — conserve them

Photon number changes only through interactions with matter (the Boltzmann equation):

$$\frac{\partial f}{\partial t} + c\,\hat{\mathbf{n}}\cdot\nabla f = \left(\frac{\partial f}{\partial t}\right)_{\text{coll}}.$$

With no matter the right side is zero: f, and hence I_ν/ν^3, is constant along a ray (Liouville's theorem, Lecture RT1).

Step 3 — write the gains and losses

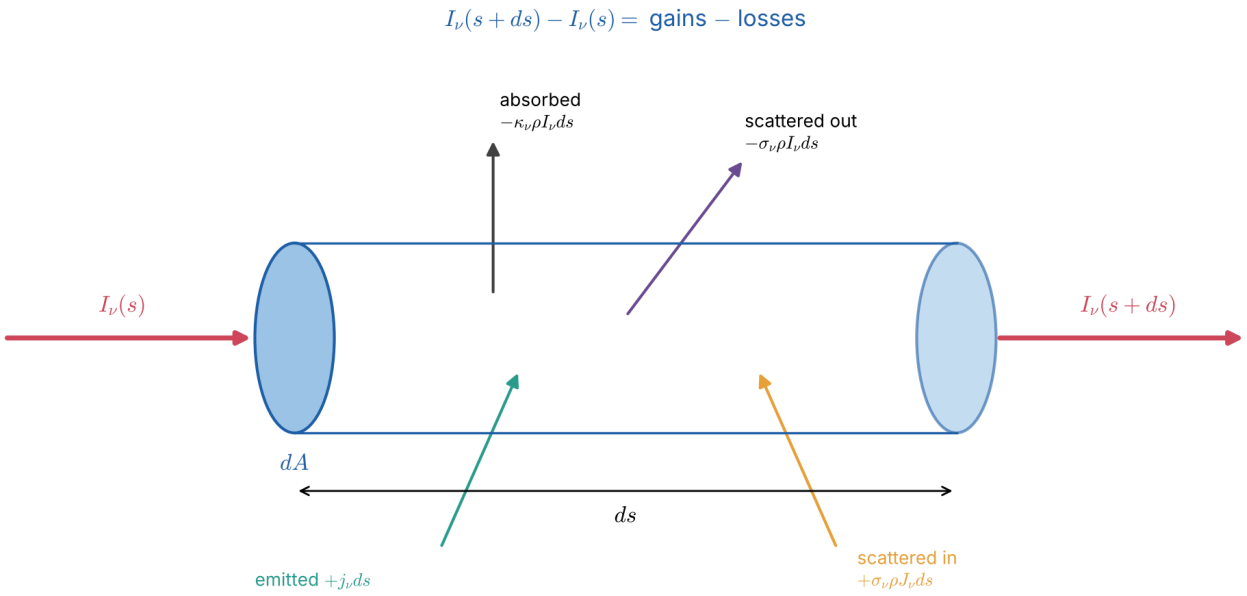


Figure 3.1 — A pencil of light crosses a small cylinder. Losses: true absorption and scattering out of the beam. Gains: thermal emission and scattering into the beam from other directions.

Process	Term in dI_ν/ds	Comment
true absorption	$-\kappa_\nu \rho I_\nu$	energy goes into the gas
stimulated emission	$+\kappa_\nu^{\text{stim}} \rho I_\nu$	$\propto I$, so it is folded into κ_ν as negative absorption
thermal emission	$+j_\nu$	in LTE $j_\nu = \kappa_\nu \rho B_\nu$ (Kirchhoff)
scattering out	$-\sigma_\nu \rho I_\nu$	photon redirected, not destroyed
scattering in	$+\frac{\sigma_\nu \rho}{4\pi} \iint R(\nu', \nu; \hat{\mathbf{n}}', \hat{\mathbf{n}})\, I_{\nu'}\, d\Omega' d\nu'$	the redistribution function R says where photons go

Together they give the **general transfer equation** (Collins eq. 9.2.23):

$$\boxed{\frac{1}{c}\frac{\partial I_\nu}{\partial t} + \hat{\mathbf{n}} \cdot \nabla I_\nu = \chi_\nu \left(S_\nu - I_\nu\right)}, \qquad \chi_\nu = (\kappa_\nu + \sigma_\nu)\rho,$$

and, for coherent isotropic scattering (where the in-scattering term is just $\sigma_\nu \rho J_\nu$), $S_\nu = \varepsilon_\nu B_\nu + (1 - \varepsilon_\nu) J_\nu$.

Why it is hard: the losses are local, but in-scattering depends on light arriving from *every* direction, so the equation is integro-differential. Most of the history of the subject (Lecture RT6) is the search for ways around this.

Step 4 — choose the geometry

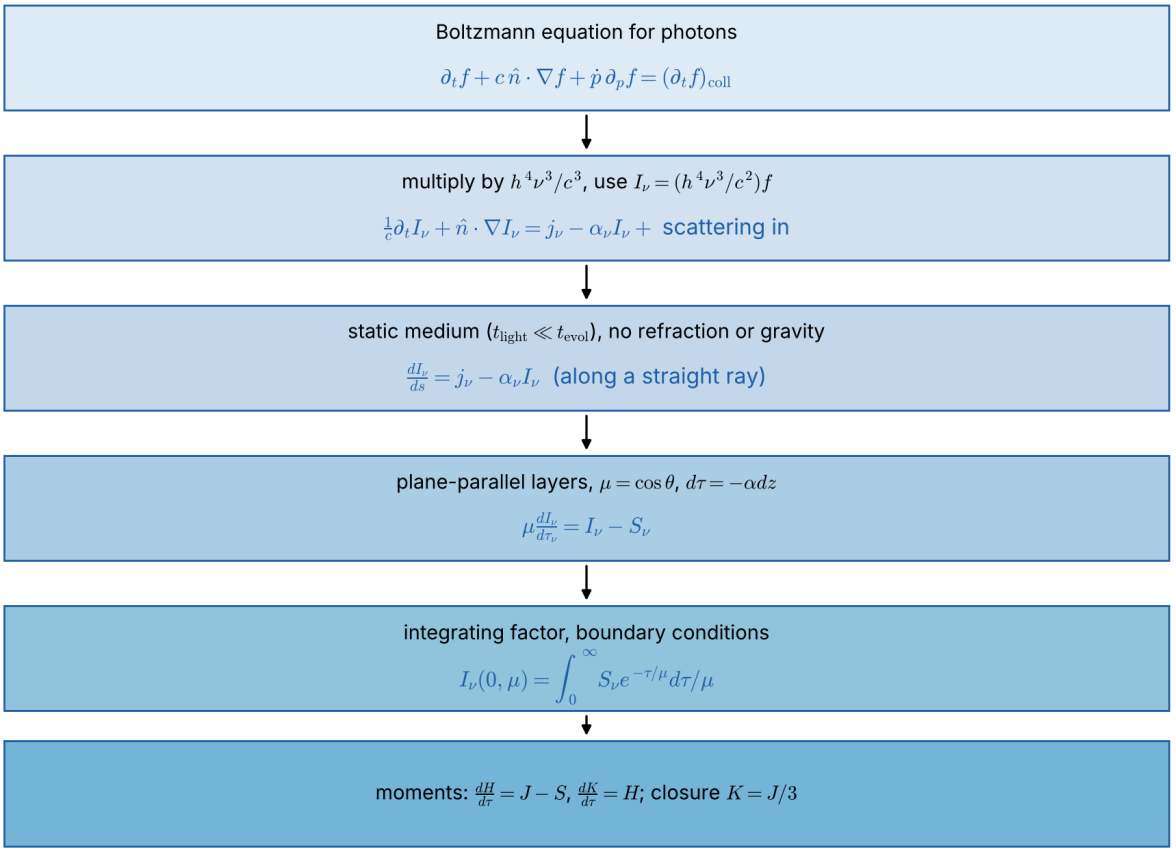


Figure 3.2 — Each simplification adds one assumption. Knowing which one was made tells you when a formula is valid.

Situation	Form	Assumption
static	$\hat{\mathbf{n}}\cdot\nabla I_\nu = \chi_\nu(S_\nu - I_\nu)$	light-crossing time ≪ evolution time
along a ray	$dI_\nu/d\tau_\nu = S_\nu - I_\nu$	$d\tau_\nu = \chi_\nu\, ds$
plane-parallel	$\mu\, dI_\nu/d\tau_\nu = I_\nu - S_\nu$	thin horizontal layers; τ measured *inward*
spherical	$\mu\,\partial_r I + \frac{1-\mu^2}{r}\partial_\mu I = \chi(S - I)$	extended atmospheres, winds
moving	comoving frame, Doppler-shifted χ_ν	winds, supernovae (Lecture SA5)

The extra spherical term exists because a straight ray changes its angle to the local radius as it travels; it vanishes when r is large compared with the atmosphere's thickness.

Energy and momentum. Integrating over angle and frequency gives $\partial_t u + \nabla\cdot\mathbf{F} = 4\pi\int(j_\nu - \kappa_\nu\rho J_\nu)\,d\nu$. If matter emits exactly what it absorbs (**radiative equilibrium**), the flux is constant: the basis of the gray atmosphere (Lecture SA4). The next moment gives the **radiation force** $\mathbf{f}_{\rm rad} = \frac{1}{c}\int\chi_\nu\mathbf{F}_\nu\,d\nu$, which drives hot-star winds.

3.2 Solving it: the formal solution

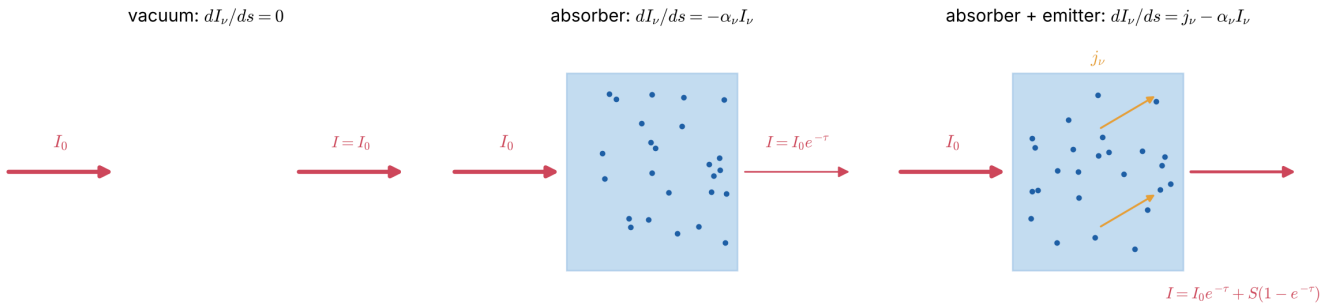


Figure 3.3 — After MIT 8.901 Fig. 9: (left) vacuum, $dI/ds = 0$; (centre) pure absorption, $I = I_0 e^{-\tau}$; (right) absorption plus emission.

Multiply $dI_\nu/d\tau_\nu + I_\nu = S_\nu$ by the integrating factor e^{τ_ν} and integrate:

$$\boxed{I_\nu(\tau_\nu) = I_\nu(0)\, e^{-\tau_\nu} + \int_0^{\tau_\nu} S_\nu(t)\, e^{-(\tau_\nu - t)}\, dt}$$

Read it as: the background dimmed by the whole column, plus light from each layer t, dimmed by the gas between it and the observer.

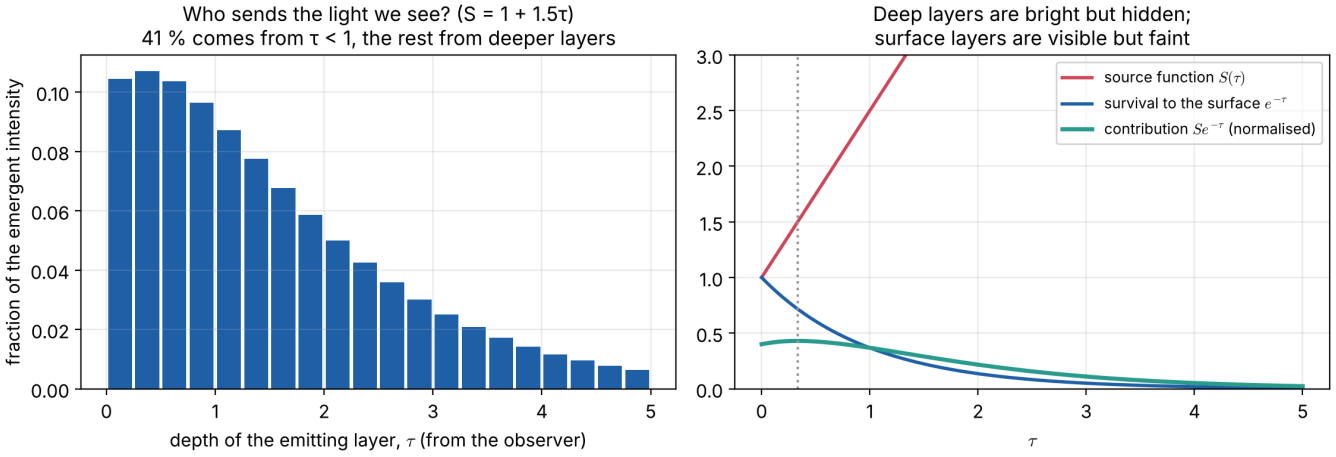


*Figure 3.4 — Who sends the light we see? For $S = 1 + 1.5\tau$, deep layers are bright but hidden, surface layers are visible but faint; their product, the **contribution function** $S\, e^{-\tau}$, peaks near $\tau \approx 1/3$. About 41 % of the emergent light comes from $\tau < 1$, and the rest from deeper.*

Why "formal"? If S_ν depends on I_ν (scattering, NLTE), this is an integral equation, not an answer. In **LTE without scattering**, $S_\nu = B_\nu(T)$ is known once $T(\tau)$ is known, and the formal solution *is* the solution.

3.3 The uniform slab and why lines exist

For constant S:

$$\boxed{I_\nu = I_\nu(0)\, e^{-\tau_\nu} + S_\nu\left(1 - e^{-\tau_\nu}\right)}$$

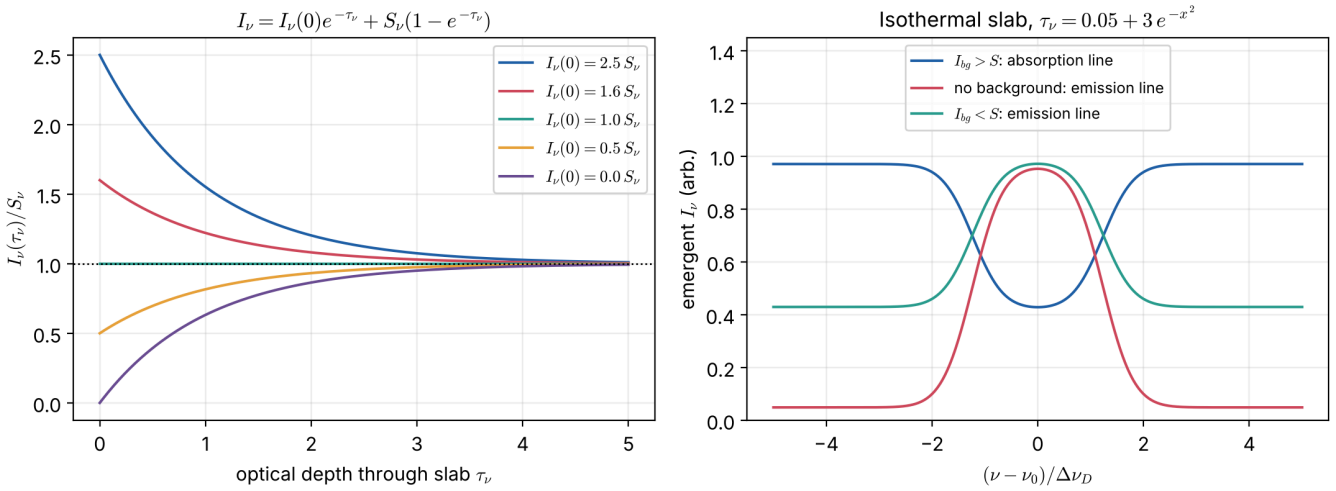


Figure 3.5 — Left: whatever the starting intensity, $I \to S$ within a few optical depths. Right: if the line centre is more opaque than the wings, a background brighter than S gives an absorption line, and a fainter one gives an emission line.

Limit	Result	Meaning
thin, $\tau \ll 1$	$I \approx I_0 + \tau(S - I_0)$	the change is proportional to the column density
thick, $\tau \gg 1$	$I \to S$	the background is forgotten
thick and LTE	$I = B_\nu(T)$	only an **isothermal** thick body radiates as a black body

Kirchhoff's three laws of spectroscopy, explained. (1) A hot, opaque body gives a continuum. (2) A thin gas on a dark sky gives $I = \tau_\nu B_\nu$: bright only where τ_ν is large, so **emission lines**. (3) Gas in front of a brighter background ($I_0 > S$) pulls the opaque frequencies down toward S: **absorption lines**. The real criterion is *background versus source function*, not "cold versus hot".

MIT's toy stars (Figure 3.6) put this to work:

- a. a hot, thin hydrogen cloud on a dark sky shows emission lines at the Rydberg wavelengths, $1/\lambda = R_\mathrm{H}(1/n_1^2 - 1/n_2^2)$;
- b. a thick base at T_H under a cooler thin layer at T_C gives $I_\nu = B_\nu(T_H)e^{-\tau_\nu} + B_\nu(T_C)(1 - e^{-\tau_\nu})$. In the continuum we see the hot base; at the lines we see the cool layer. The result is absorption lines;
- c. a **thermal inversion** ($T_C > T_H$) turns the same lines into emission lines, as in the solar chromosphere, planetary stratospheres and hot Jupiters.

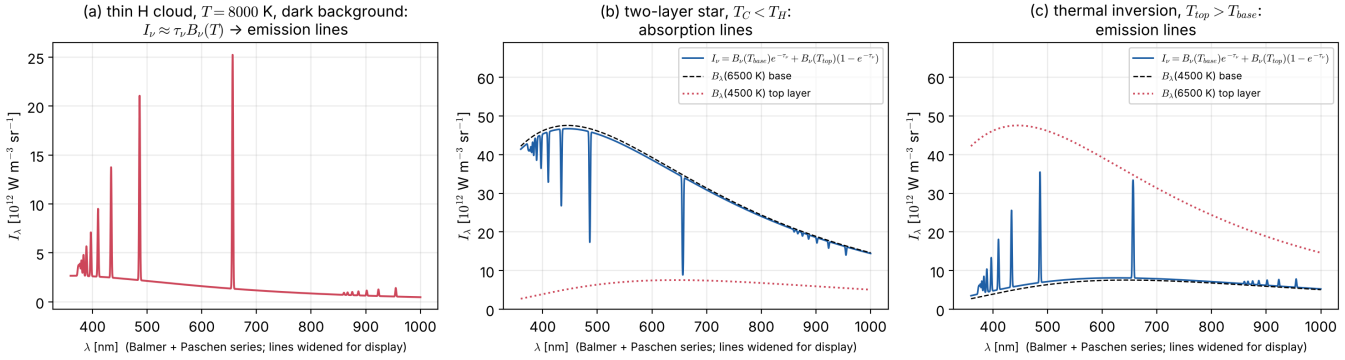


Figure 3.6 — The MIT 8.901 toy models computed exactly: (a) thin 8000 K cloud, emission lines; (b) 6500 K base under a 4500 K layer, absorption lines; (c) inverted case, emission lines. Line widths are exaggerated.

Radio H II regions. For free–free absorption, $T_b = T_e(1 - e^{-\tau_\nu})$ with $\tau_\nu \propto \nu^{-2.1}$. With $T_e = 8000$ K, T_b is 80, 761, 5057, 7602 and 8000 K for τ = 0.01, 0.1, 1, 3 and 10. The spectrum is a black body ($\propto \nu^2$) at low frequency and almost flat ($\propto \nu^{-0.1}$) at high frequency, with the turnover at $\tau \approx 1$.

Worked example 3.1 — The two-layer star at Hα (MIT model)

Base $T_H = 6500$ K, layer $T_C = 4500$ K. At 656.3 nm the layer has $\tau = 0.02$ in the continuum and $\tau = 2.02$ at line centre. Find the line depth, test the linearised formula, and find the maximum possible depth.

Solution. $B_\lambda(6500) = 3.474 \times 10^{13}$ and $B_\lambda(4500) = 7.551 \times 10^{12}$ W m^{-3} sr^{-1}.

- Continuum: $I_c = B_H e^{-0.02} + B_C(1 - e^{-0.02}) = 0.9845\, B_H$. Line: $I_l = B_H e^{-2.02} + B_C(1 - e^{-2.02}) = 0.321\, B_H$. **Depth** $= 1 - I_l/I_c = 0.67$.
- The linear form $B_H - \tau(B_H - B_C)$ is excellent at $\tau = 0.02$ but gives $-0.57\, B_H$ at $\tau = 2$, which is negative and meaningless. Strong lines need the exponential.
- As $\tau \to \infty$, $I_l \to B_C$, so the deepest possible line has depth $1 - B_C/I_c = 0.78$. **A saturated line's depth measures the temperature contrast of the atmosphere.**

Worked example 3.2 — A linear source function: the seed of Eddington–Barbier

$S(t) = a + bt$, $I(0) = 0$, total optical depth τ. Find $I(\tau)$ and its thick limit.

Solution. With $u = \tau - t$:

$$I = \int_0^\tau [a + b(\tau - u)]e^{-u}du = (a + b\tau)(1 - e^{-\tau}) - b[1 - (1 + \tau)e^{-\tau}] \xrightarrow{\tau\to\infty} a + b(\tau - 1).$$

The emerging intensity equals the source function one optical depth back from the observer. Lecture RT4 turns this into the Eddington–Barbier relation.

Worked example 3.3 — When does time matter?

Compare photon escape and evolution times for (a) the solar photosphere and (b) a supernova 10 days after explosion ($v = 10^4$ km s^{-1}, $\tau \approx 100$).

Solution.

- a. Photons cross the ~500 km photosphere in ~2 ms, while granulation changes in ~5 minutes. The **static** equation is excellent.
- b. $R = vt = 8.6 \times 10^{12}$ m, and the diffusion time is $t_d \approx \tau R/c \approx 2.9 \times 10^6$ s ≈ 33 days, longer than the age. The **time-dependent** equation is needed, which is why supernova light curves peak late.

Common misconceptions

1. **“The RTE is an empirical law.”** It follows from photon conservation. Only the coefficients κ_ν, σ_ν, j_ν and R need microphysics.
2. **“The formal solution solves the problem.”** Only when S_ν is known in advance. With scattering, S depends on I.
3. **“An optically thick object always emits a black body.”** Only if it is also isothermal. A star is thick but not isothermal, so its spectrum is not exactly Planckian.
4. **“Emission lines need hot gas.”** They need S_ν larger than the background: thin gas on a dark sky, or temperature rising outward.

Lecture summary

- Photon conservation plus five gain and loss terms gives the general transfer equation, $\frac{1}{c}\partial_t I + \hat{\mathbf{n}} \cdot \nabla I = \chi(S - I)$.
- Each special form (static, ray, plane-parallel, spherical) adds one assumption.
- Formal solution: background dimmed by $e^{-\tau}$, plus each layer’s emission dimmed by the gas in front; we see about one optical depth in.
- Slab: thin, $I \approx I_0 + \tau(S - I_0)$; thick, $I \to S$.
- Absorption lines need background $> S$; emission lines need the reverse. The MIT models (cloud, two-layer star, inversion) show all three cases.

Exam questions and answers — Lecture RT3

Q1. Write the general time-dependent transfer equation with LTE thermal emission and coherent isotropic scattering. Name each term, and say which term makes the equation hard to solve.

> **A1.** $\frac{1}{c}\partial_t I_\nu + \hat{\mathbf{n}} \cdot \nabla I_\nu = -\kappa_\nu \rho I_\nu$ (absorption, corrected for stimulated emission) $- \sigma_\nu \rho I_\nu$ (scattering out) $+ \kappa_\nu \rho B_\nu$ (thermal emission, Kirchhoff) $+ \sigma_\nu \rho J_\nu$ (scattering in). The scattering-in term couples all directions through J_ν, which makes the equation integro-differential and non-local.

Q2. Starting from $dI_\nu/ds = -\alpha_\nu I_\nu + j_\nu$, derive the formal solution and the uniform-slab result. Discuss the thin and thick limits.

> **A2.** With $d\tau = \alpha\, ds$ and $S = j/\alpha$: $dI/d\tau + I = S$. Multiply by e^τ: $d(Ie^\tau)/d\tau = Se^\tau$, so $I(\tau) = I(0)e^{-\tau} + \int_0^\tau S(t)e^{-(\tau - t)}dt$. For constant S, $\int_0^\tau e^{-(\tau-t)}dt = 1 - e^{-\tau}$. Thin: $I \approx I_0 + \tau(S - I_0)$, linear in the column density. Thick: $I \to S$, independent of I_0.

Q3. “Absorption lines form because the gas in a stellar atmosphere is cooler than the gas below it.” Rephrase this precisely using I_ν and S_ν.

> **A3.** At line frequencies τ_ν is larger, so the emergent intensity is set by S_ν in *higher* layers. If S_ν decreases outward (in LTE, $S_\nu = B_\nu(T)$ with T falling outward), line photons come from layers of smaller S_ν than

continuum photons: an absorption line. If S_ν increases outward (a chromosphere, or NLTE effects), the line is in emission.

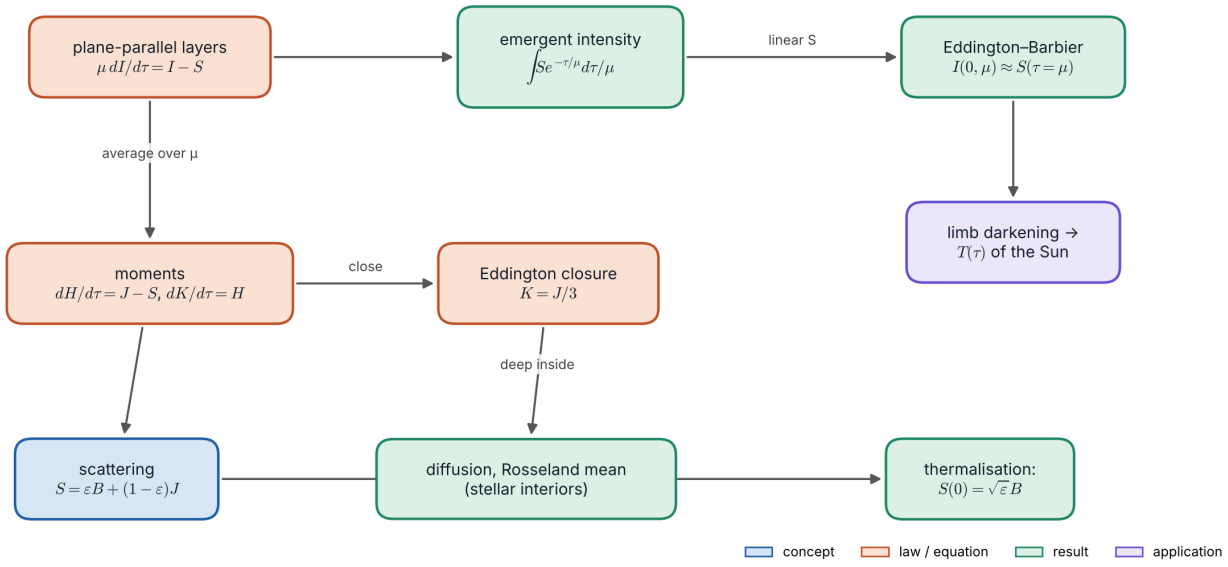
Q4. (MIT 8.901.) A star has an optically thick base at T_H and a thin layer at T_C. Show that $I_\nu \approx B_\nu(T_H) - \tau_\nu[B_\nu(T_H) - B_\nu(T_C)]$ for $\tau_\nu \ll 1$, explain the absorption lines, and say what changes in a thermal inversion.

A4. The thick base gives $I_\nu(0) = B_\nu(T_H)$; the layer has $S_\nu = B_\nu(T_C)$. So $I_\nu = B_\nu(T_H)e^{-\tau} + B_\nu(T_C)(1 - e^{-\tau})$, and $e^{-\tau} \approx 1 - \tau$ gives the result. Where α_ν is large (the lines), the intensity drops toward $B_\nu(T_C) < B_\nu(T_H)$: absorption lines. In an inversion $T_C > T_H$, so the same lines appear in emission. For strong lines the linear form fails and the full exponential is needed (Worked example 3.1).

Lecture RT4 — Atmospheres: Plane-Parallel Layers, Moments and Scattering

> **Sources:** Collins Ch. 9.2–9.4 (plane-parallel RTE, moments, radiative equilibrium, Eddington approximation); Chandrasekhar Ch. I–III (Schwarzschild–Milne equations, Λ operator); Hu, *Astro 305 Set 1* (diffusion, Rosseland mean); Novotny Ch. 2.4–2.5 (solution of the equation of transfer, solar examples).

RT4 concept map — atmospheres, moments and scattering



*Concept map — the top row is about **what we see**: in plane-parallel layers the emergent intensity is a weighted average of S, and Eddington–Barbier says where we look. The bottom rows are about **what is inside**: moments need a closure, deep layers behave like diffusion, and scattering makes S fall below B near the surface.*

Learning goals

1. Write the plane-parallel equation and its emergent-intensity solution.
2. Derive and use the Eddington–Barbier relations, and read temperatures from limb darkening.
3. Form the moment equations, state radiative equilibrium, and apply the Eddington closure.
4. Derive the diffusion approximation and the Rosseland mean.
5. Explain the $\sqrt{\varepsilon}$ law, the thermalisation depth, and why Λ-iteration fails.

The big ideas in plain words

A star's atmosphere is a stack of thin layers. It is thin compared with the star's radius, so each layer is a flat sheet. Only depth matters, and the angle θ of a ray to the vertical, through $\mu = \cos\theta$.

We see down to about one optical depth *along the line of sight*. Looking straight down (disk centre) we reach vertical depth $\tau \approx 1$; looking obliquely (near the limb) we reach only $\tau \approx \mu$, higher up. If the temperature falls outward, the

limb looks darker. Limb darkening is a thermometer that reads temperature at different heights.

The flux comes from $\tau \approx 2/3$. Averaging over the disk, the light leaving a star represents conditions at $\tau \approx 2/3$. That is where the "photosphere" is, and why $T(2/3) \approx T_{\rm eff}$.

Deep inside, light diffuses like heat. Many mean free paths down, the radiation is almost isotropic with a tiny excess outward, proportional to the temperature gradient. Energy then leaks out like heat through a wall, mostly through the frequency "windows" where opacity is low: the **Rosseland mean**.

Scattering lets photons leak out from deep layers. If a photon is destroyed only with probability ε per interaction, it can random-walk about $1/\sqrt{\varepsilon}$ optical depths before thermalising. Within that depth of the surface the light "feels" the open boundary, and S falls to $\sqrt{\varepsilon}\,B$ at the top even if the gas is isothermal.

4.1 The plane-parallel equation and what we see

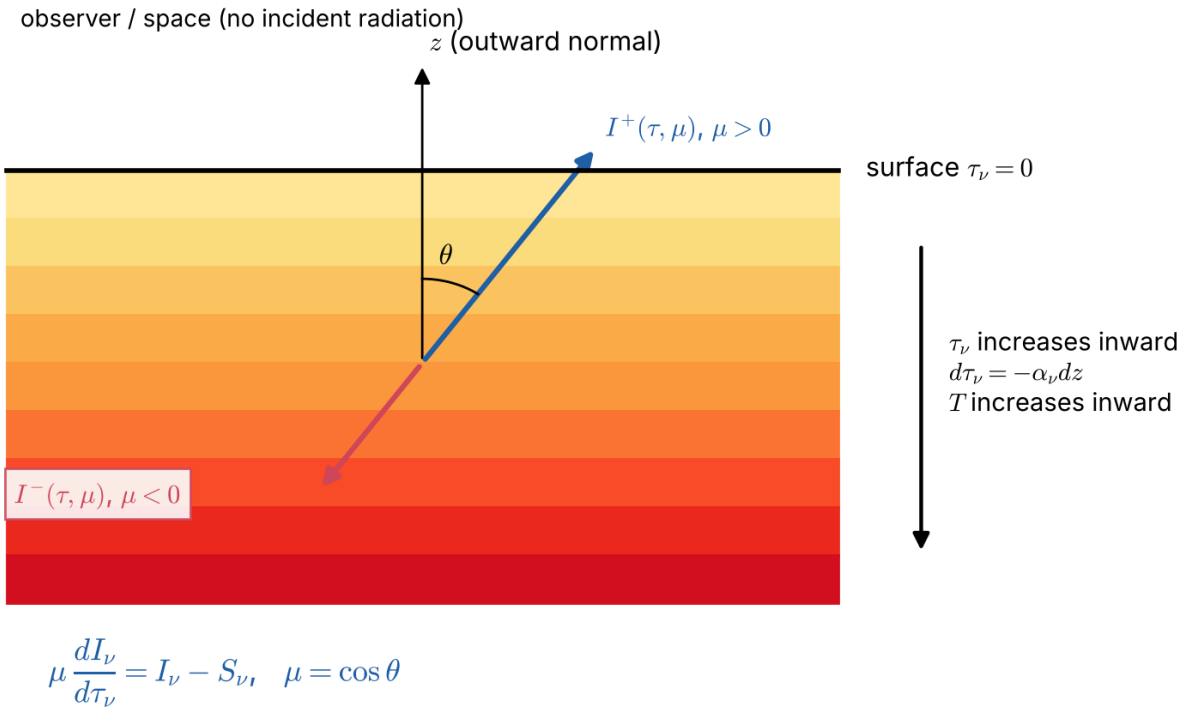


*Figure 4.1 — z points outward, $\mu = \cos\theta$, outgoing rays have $\mu > 0$. Optical depth increases **inward**: $d\tau_\nu = -\alpha_\nu dz$.*

$$\boxed{\mu \frac{dI_\nu}{d\tau_\nu} = I_\nu - S_\nu}$$

with no light entering from above ($I(0, \mu < 0) = 0$). Integrating outward from great depth gives the emergent intensity:

$$\boxed{I_\nu(0,\mu) = \int_0^\infty S_\nu(t)\, e^{-t/\mu}\, \frac{dt}{\mu}}$$

This is a **Laplace transform** of $S_\nu(\tau)$: scanning the disk from centre ($\mu = 1$) to limb ($\mu \to 0$) samples S at different depths.

Schwarzschild–Milne equations. Inserting the formal solution into the moments gives, with exponential integrals $E_n(x) = \int_1^\infty e^{-xt} t^{-n} dt$:

$$J_\nu(\tau) = \tfrac{1}{2}\int_0^\infty S_\nu(t) E_1|t-\tau|\, dt \equiv \Lambda_\tau[S_\nu], \qquad F_\nu(\tau) = 2\pi\Big[\int_\tau^\infty S E_2(t-\tau)dt - \int_0^\tau S E_2(\tau - t$$

The **lambda operator** Λ turns a source function into a mean intensity. Example: constant S gives $J(0) = S/2$ (half the sky is empty) and $J \to S$ deep down.

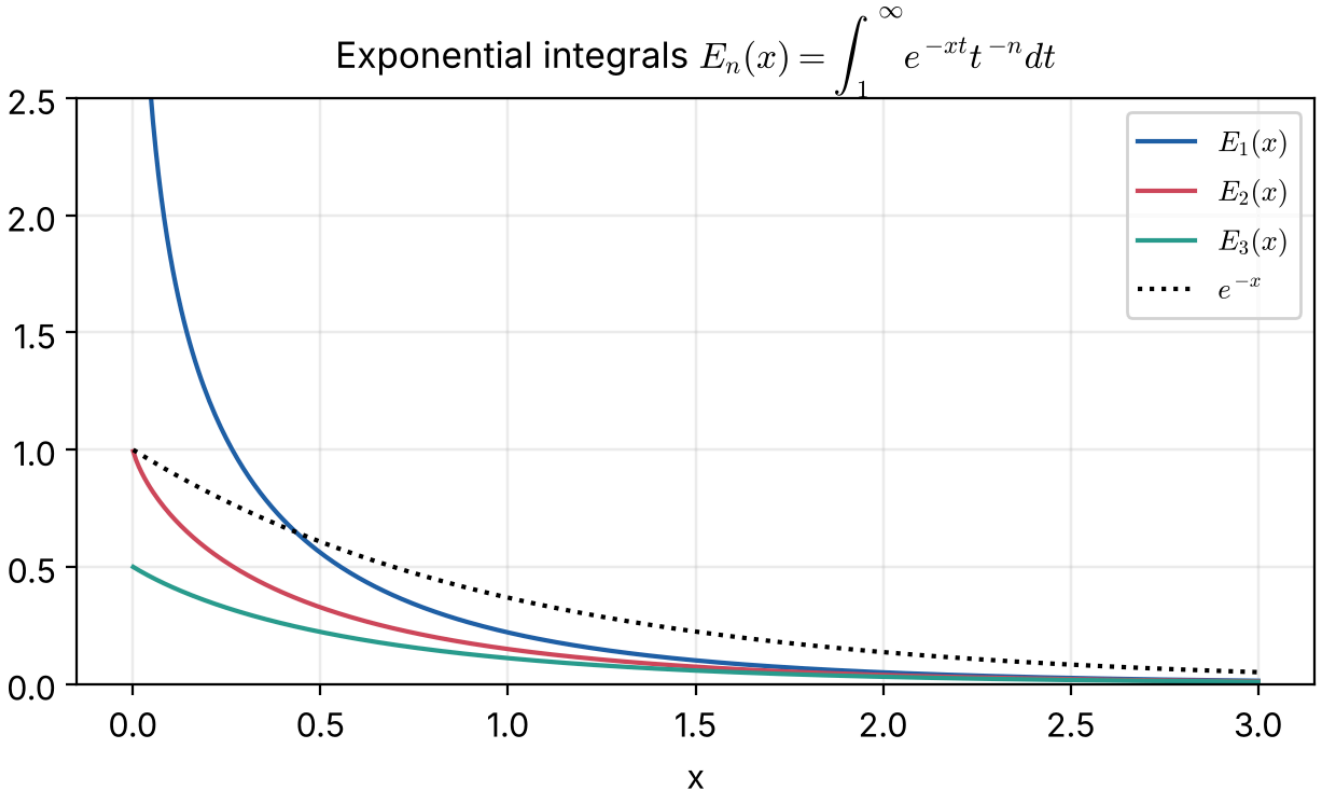


Figure 4.2 — E_1 diverges logarithmically at 0 but is integrable; all E_n decay roughly as e^{-x}/x. $E_n(0) = 1/(n-1)$.

4.2 Eddington–Barbier: where we look

For $S = a + b\tau$, using $\int_0^\infty e^{-t/\mu} dt/\mu = 1$ and $\int_0^\infty t e^{-t/\mu} dt/\mu = \mu$:

$$\boxed{I_\nu(0,\mu) = S_\nu(\tau_\nu = \mu)} \qquad \boxed{F_\nu(0) = \pi S_\nu(\tau_\nu = 2/3)}$$

These are exact for linear S and approximate otherwise; a quadratic term $c\tau^2$ adds an error $c\mu^2$.

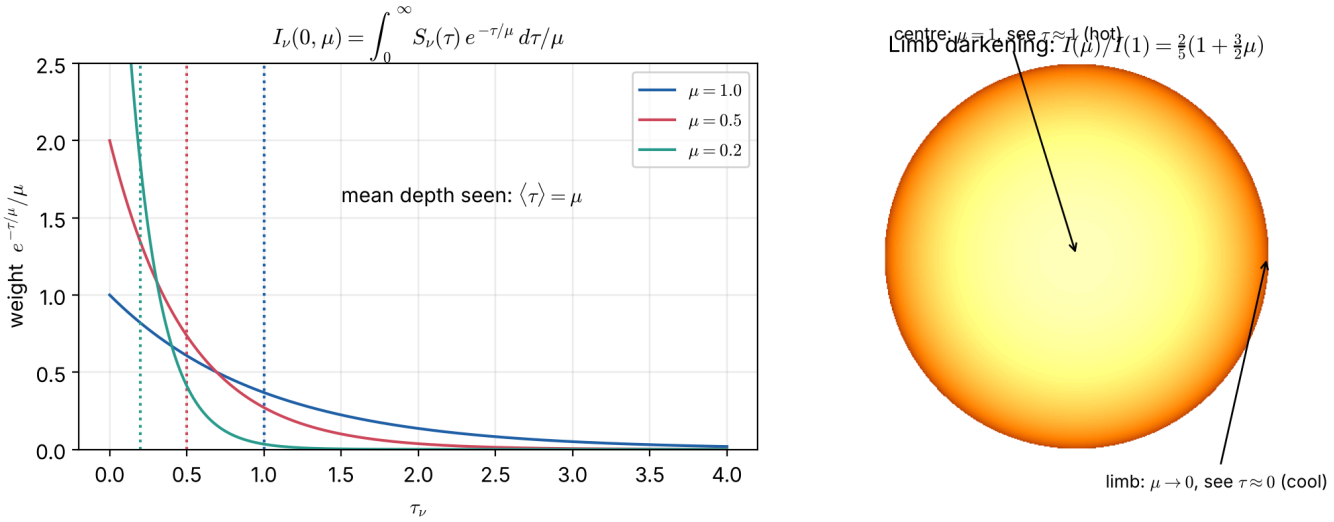


Figure 4.3 — Left: the weighting kernel $e^{-\tau/\mu}/\mu$ has mean depth μ. Right: limb darkening. At disk centre we see down to $\tau \approx 1$ (hot); near the limb we see higher layers (cool).

Limb darkening as a thermometer. In LTE, $S_\nu = B_\nu(T)$. Measure $I(0,\mu)$, read off $S(\tau = \mu)$, and invert the Planck function:

$$T = \frac{hc/\lambda k}{\ln\!\left(1 + 2hc^2/\lambda^5 S\right)}.$$

This is how the first empirical models of the solar photosphere were built (Worked example 4.1).

4.3 Moments, radiative equilibrium and the Eddington closure

Averaging the plane-parallel equation over μ, with weights 1 and μ:

$$\boxed{\frac{dH_\nu}{d\tau_\nu} = J_\nu - S_\nu, \qquad \frac{dK_\nu}{d\tau_\nu} = H_\nu}$$

Each equation brings in a higher moment ($J \to H \to K \to \ldots$). This **closure problem** must be cut somewhere.

Radiative equilibrium. If radiation carries all the energy, the flux is constant and energy absorbed equals energy emitted:

$$\boxed{\int_0^\infty \kappa_\nu J_\nu \, d\nu = \int_0^\infty \kappa_\nu B_\nu(T)\, d\nu}$$

Coherent scattering drops out, because it moves energy without destroying it. **This condition fixes $T(\tau)$**: it is what turns the transfer equation into a model atmosphere (Lecture SA4).

The Eddington approximation. Assume $K = J/3$ everywhere. This is exact for isotropic or linear-in-μ fields (Lecture RT1), so it is good at depth. At the surface, assuming isotropic outgoing light gives

$$\boxed{J(0) = 2H(0)}$$

Combined, $\frac{1}{3}\, d^2 J_\nu/d\tau_\nu^2 = J_\nu - S_\nu$: a simple second-order ODE. It is typically good to a few per cent.

4.4 Deep inside: diffusion and the Rosseland mean

Where $S = B$ varies slowly over a mean free path, $I_\nu \approx B_\nu + \mu\, dB_\nu/d\tau_\nu$, so

$$F_\nu = \frac{4\pi}{3}\frac{dB_\nu}{d\tau_\nu} = -\frac{4\pi}{3\alpha_\nu}\frac{\partial B_\nu}{\partial T}\frac{dT}{dz}.$$

Summing over frequency adds the "conductances" $1/\alpha_\nu$, like resistors in parallel:

$$\boxed{\frac{1}{\kappa_R} = \frac{\int \kappa_\nu^{-1}\, \partial B_\nu/\partial T\, d\nu}{\int \partial B_\nu/\partial T\, d\nu}} \quad \Rightarrow \quad \boxed{F = -\frac{16\sigma T^3}{3\kappa_R \rho}\frac{dT}{dz}}$$

This is radiative transfer in **stellar interiors**. As a harmonic mean, the Rosseland mean is dominated by the most transparent frequencies.

4.5 Scattering and thermalisation

With $S = \varepsilon B + (1-\varepsilon)J$ in an **isothermal** atmosphere, the Eddington equation becomes $\frac{1}{3}J'' = \varepsilon(J - B)$, whose solution is

$$J(\tau) = B\Big[1 - \frac{e^{-\sqrt{3\varepsilon}\,\tau}}{1 + 2\sqrt{\varepsilon/3}}\Big], \qquad S(0) \approx 1.15\sqrt{\varepsilon}\, B.$$

The exact theory (Chandrasekhar's H-functions) gives the famous

$$\boxed{S(0) = \sqrt{\varepsilon}\, B}, \qquad \tau_{\rm th} \sim \frac{1}{\sqrt{\varepsilon}}.$$

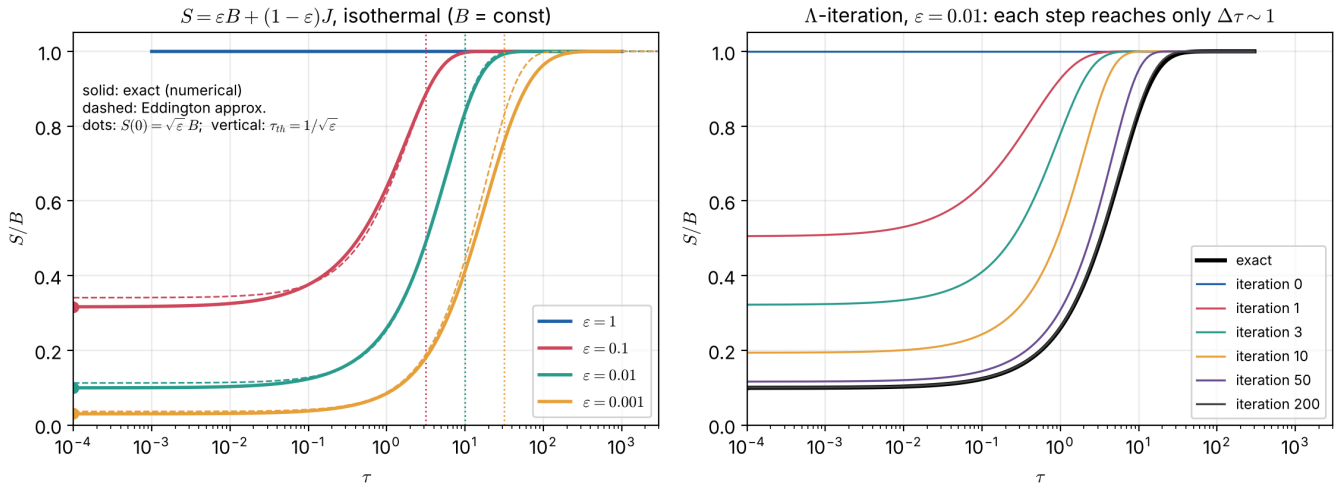


Figure 4.4 — Left: exact solutions (solid, from `scripts/lte_solvers.py`) and the Eddington approximation (dashed). Although B is constant, S falls far below B near the surface and reaches B only below $\tau_{\rm th}$ (dotted). Right: Λ-iteration for $\varepsilon = 0.01$ after up to 200 steps, still far from converged.

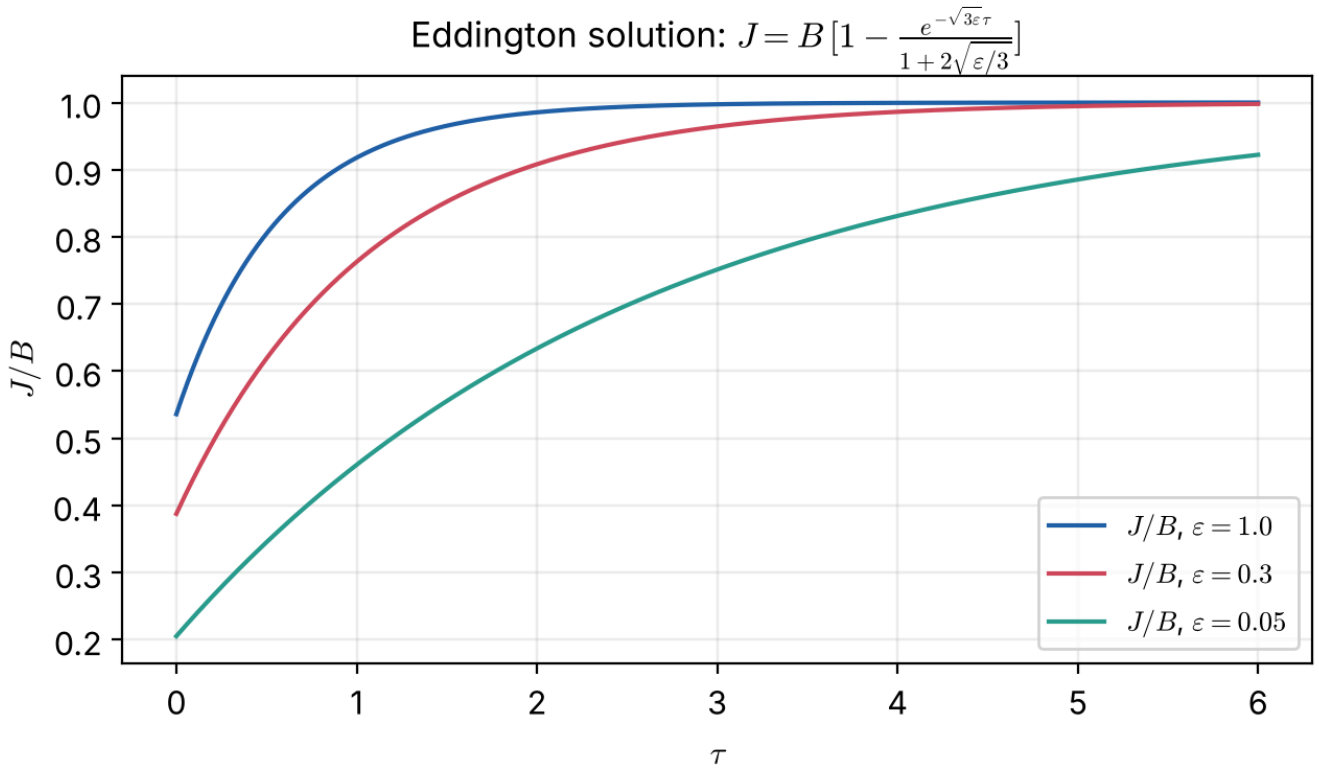


Figure 4.5 — The Eddington solution J/B for three values of ε. The smaller ε, the deeper the radiation "feels" the open surface: the deficit $J < B$ reaches down to $\tau \sim 1/\sqrt{3\varepsilon}$.

Why it matters. This is NLTE behaviour in a gas whose temperature is perfectly uniform: escape alone drives S below B. For strong resonance lines $\varepsilon \sim 10^{-4}$–10^{-8}, so thermalisation happens hundreds to thousands of optical depths down, and line cores are far darker than LTE predicts (Lecture SA3).

Why Λ-iteration fails. The obvious scheme $S^{(n+1)} = \varepsilon B + (1-\varepsilon)\Lambda[S^{(n)}]$ spreads information only about one optical depth per step, and its error shrinks only as $(1-\varepsilon)^n$. It can *look* converged while being badly wrong. Our script instead builds Λ as a matrix (piecewise-linear S, analytic E_n integrals) and solves $(\mathbf{1} - (1-\varepsilon)\mathbf{\Lambda})\mathbf{S} = \varepsilon\mathbf{B}$ directly. Modern codes use **accelerated Λ-iteration (ALI)**.

Worked example 4.1 — The Sun's temperature from limb darkening

At 500 nm, $I(\mu)/I(1) = 0.36 + 0.64\mu$ and $I(1) = 4.0 \times 10^{13}$ W m^{-3} sr^{-1}. Assuming LTE, find T at τ_{500} = 0, 0.5 and 1.

Solution. By Eddington–Barbier, $S(\tau) = I(1)(0.36 + 0.64\tau)$. Invert the Planck function with $hc/\lambda k = 28\,776$ K and $2hc^2/\lambda^5 = 3.811 \times 10^{15}$:

τ_{500}	$S/I(1)$	T [K]
0	0.36	5155
0.5	0.68	5814
1	1.00	6300

The temperature rises inward by about 1100 K over one optical depth. This is observational proof that the Sun is not isothermal, and the reason it shows absorption lines.

Worked example 4.2 — Thermalisation of a resonance line

A chromospheric resonance line has $\varepsilon = 10^{-4}$. Estimate $S(0)/B$, the thermalisation depth and the number of scatterings.

Solution. $S(0) = \sqrt{\varepsilon}\,B = 0.01\,B$; $\tau_{\rm th} \approx 1/\sqrt{\varepsilon} = 100$ (Eddington: $1/\sqrt{3\varepsilon} = 58$); $N \approx 1/\varepsilon = 10^4$. An LTE model would make the line core 100 times too bright, which is why Ca II H & K and Mg II h & k need NLTE.

Worked example 4.3 — Rosseland versus straight mean

Split $\partial B_\nu/\partial T$ equally between two bands with $\kappa_1 = 1$ and $\kappa_2 = 100$.

Solution. $1/\kappa_R = \tfrac{1}{2}(1 + 0.01) = 0.505$, so $\kappa_R = 1.98$, while the straight mean is 50.5. The flux flows through the transparent band. Line blanketing raises κ_R only modestly unless the lines fill the windows.

Common misconceptions

1. **"The limb is darker because it is farther away."** The extra distance is negligible; we see higher, cooler layers there.
2. **"Eddington–Barbier is exact."** Only for a source function linear in τ.
3. **"An isothermal atmosphere has $S = B$ everywhere."** With scattering, S falls to $\sqrt{\varepsilon}B$ at the surface.
4. **"If an iteration changes little, it has converged."** Λ-iteration can creep along with tiny steps while still far from the answer.

Lecture summary

- Plane-parallel: $\mu\,dI/d\tau = I - S$; the emergent intensity is a Laplace transform of S.
- Eddington–Barbier: $I(0,\mu) \approx S(\mu)$ and $F \approx \pi S(2/3)$; limb darkening measures $T(\tau)$.
- Moments $dH/d\tau = J - S$, $dK/d\tau = H$ need a closure: $K = J/3$ and $J(0) = 2H(0)$.
- Radiative equilibrium, $\int \kappa_\nu (J_\nu - B_\nu) d\nu = 0$, fixes $T(\tau)$.
- Deep inside: diffusion with the Rosseland (harmonic) mean.
- Scattering: $S(0) = \sqrt{\varepsilon}B$, $\tau_{\rm th} \sim 1/\sqrt{\varepsilon}$, and Λ-iteration is too slow.

Exam questions and answers — Lecture RT4

Q1. Prove the Eddington–Barbier relations $I(0,\mu) = S(\tau = \mu)$ and $F(0) = \pi S(\tau = 2/3)$ for $S = a + b\tau$, and explain limb darkening. When would we see limb *brightening*?

> **A1.** $I(0,\mu) = \int_0^\infty (a + bt) e^{-t/\mu} dt/\mu = a + b\mu = S(\mu)$; $F = 2\pi \int_0^1 (a + b\mu)\mu\, d\mu = \pi(a + 2b/3) = \pi S(2/3)$. Near the limb the line of sight reaches only vertical depth μ, in higher layers; if S (in LTE, T) falls outward, the limb is darker. Where S rises outward (the chromosphere in the UV, EUV and radio), the limb is brighter.

Q2. State radiative equilibrium in integral form. Why do scattering terms drop out, and what does the condition determine?

A2. $\int \alpha_\nu (J_\nu - S_\nu) d\nu = 0$, equivalently $dF/dz = 0$. For coherent scattering that part is $\alpha^{\mathrm{sc}}(J - J) = 0$: scattering redirects energy without creating or destroying it. What remains, $\int \kappa_\nu (J_\nu - B_\nu) d\nu = 0$, determines the temperature structure $T(\tau)$.

Q3. Derive the diffusion approximation and the Rosseland mean. Why is it a harmonic mean?

A3. Deep inside, $I \approx B + \mu\, dB/d\tau$, so $F_\nu = \frac{4\pi}{3} dB_\nu/d\tau_\nu = -\frac{4\pi}{3\alpha_\nu} \frac{\partial B_\nu}{\partial T} \frac{dT}{dz}$. Summing over frequency adds the $1/\alpha_\nu$, like conductances in parallel. This gives $1/\kappa_R$ as the $\partial B_\nu/\partial T$-weighted mean of $1/\kappa_\nu$, and $F = -\frac{16\sigma T^3}{3\kappa_R \rho} \frac{dT}{dz}$.

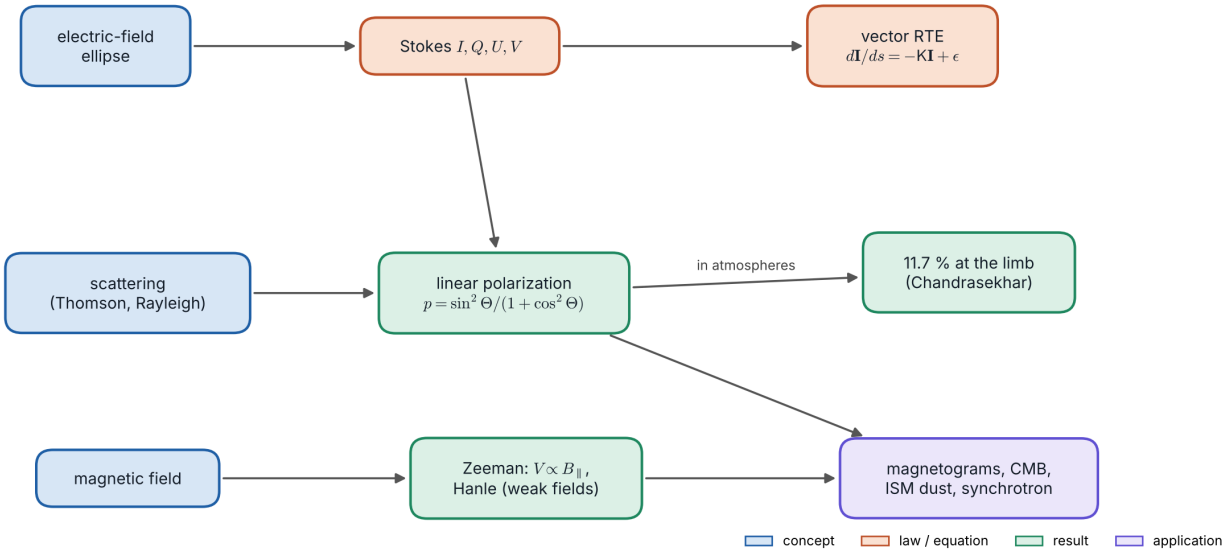
Q4. For an isothermal atmosphere with $S = \varepsilon B + (1-\varepsilon) J$, explain physically why $S < B$ near the surface, estimate the depth where $S \to B$, and explain why Λ-iteration converges slowly.

A4. Near the surface photons escape, so $J < B$; scattering ties S to J, so $S < B$ too, reaching $\sqrt{\varepsilon} B$ at $\tau = 0$. A photon survives about $1/\varepsilon$ scatterings and random-walks $\sim 1/\sqrt{\varepsilon}$ optical depths, so $S \to B$ below $\tau_{\mathrm{th}} \sim 1/\sqrt{\varepsilon}$. Λ couples points only over $\Delta\tau \sim 1$, so each iteration moves information by one optical depth, and the error decays as $(1-\varepsilon)^n$; about $1/\varepsilon$ iterations are needed. The remedy is direct solution (Feautrier, matrix inversion) or ALI.

Lecture RT5 — Polarized Light

> **Sources:** Hu, *Astro 305 Set 1* (Stokes parameters as differences of filtered intensities); Chandrasekhar, *Radiative Transfer* (1950), Ch. I §15–17 (Stokes parameters, Rayleigh phase matrix) and Ch. X §68 (exact laws of darkening in the two states of polarization; Table XXIV); Landi Degl'Innocenti & Landolfi, *Polarization in Spectral Lines* (2004); Stenflo, *Solar Magnetic Fields* (1994).

RT5 concept map — polarized light



Concept map — polarization is described by four Stokes numbers. Two physical agents create it: ***scattering*** *(left) gives linear polarization, up to 11.7 % at the limb of an electron-scattering star;* ***magnetic fields*** *(bottom) give Zeeman and Hanle signatures. Transporting the four numbers needs the vector transfer equation.*

Learning goals

1. Define the Stokes parameters I, Q, U, V and the degree and angle of polarization.
2. Write the polarized transfer equation in matrix form and name its terms.
3. Explain polarization by scattering (Thomson, Rayleigh) and Chandrasekhar's result for electron-scattering atmospheres.
4. Explain how the Zeeman and Hanle effects let us measure magnetic fields, and estimate Zeeman splittings.

The big ideas in plain words

Light is a transverse wave. The electric field oscillates perpendicular to the direction of travel. If its oscillation has a preferred orientation, the light is *polarized*. Natural thermal emission is unpolarized because countless independent emitters oscillate in random directions. Polarization appears when something breaks the symmetry: a scattering geometry, a magnetic field, aligned dust grains, or relativistic electrons spiralling in a field.

Four numbers describe any beam. Hu's picture is the clearest: put filters in front of a detector. I is the total intensity. Q is the difference between light passing a horizontal and a vertical polarizer. U is the same difference for polarizers rotated by 45°. V is the difference between right- and left-circular filters. These four Stokes parameters all have units of intensity, so they can be transported exactly like I_ν.

Why scattering polarizes. An electron hit by light oscillates along the incoming electric field and re-radiates like a small antenna, which emits nothing along its axis of oscillation. Light scattered at 90° therefore contains only one direction of oscillation: it is 100 % linearly polarized. This is why the blue sky is polarized and why the limb of a hot star should be slightly polarized.

Magnetic fields leave fingerprints. In a magnetic field, atomic levels split (the Zeeman effect) into components with definite polarization. A field along the line of sight produces *circular* polarization (V); a field across the line of sight produces *linear* polarization (Q, U). Measuring the Stokes profiles of spectral lines is how solar magnetograms are made. Very weak fields instead modify scattering polarization (the Hanle effect).

5.1 Stokes parameters

For a monochromatic wave $E_x = a_1\cos(\omega t - \delta_1)$, $E_y = a_2\cos(\omega t - \delta_2)$, with $\delta = \delta_2 - \delta_1$ (up to a common constant):

$$I = a_1^2 + a_2^2, \quad Q = a_1^2 - a_2^2, \quad U = 2a_1a_2\cos\delta, \quad V = 2a_1a_2\sin\delta.$$

Operationally (Hu): $I = I_1 + I_2$, $Q = I_1 - I_2$, $U = I_{1'} - I_{2'}$ (axes rotated by 45°), $V = I_+ - I_-$.

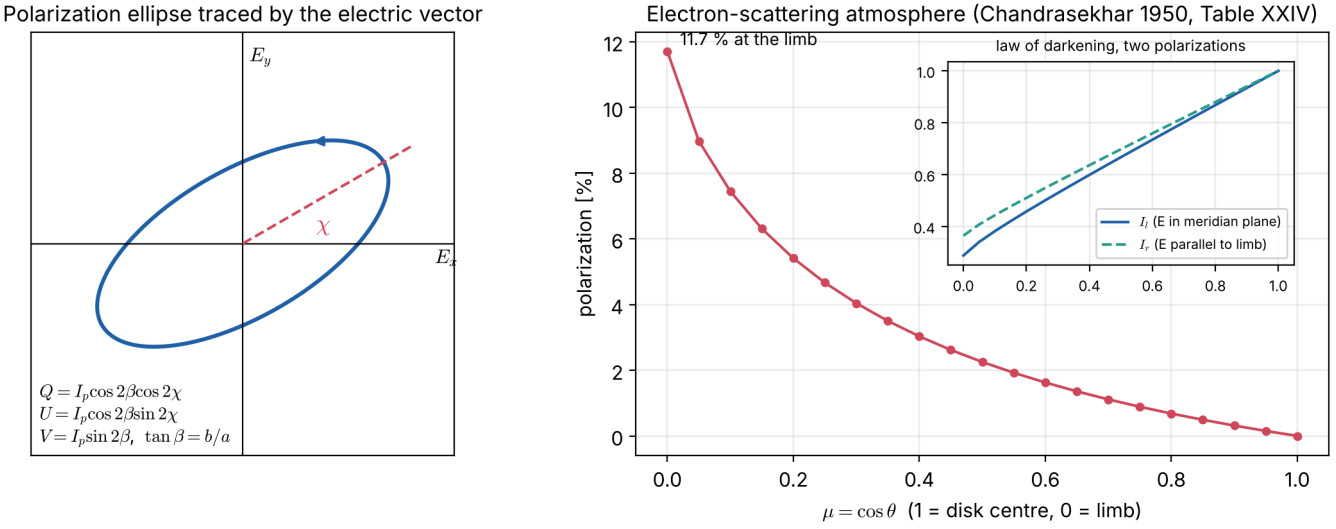


Figure 5.1 — Left: the electric vector traces an ellipse with orientation χ and ellipticity $\tan\beta = b/a$. Right: Chandrasekhar's exact result for a semi-infinite electron-scattering atmosphere. The polarization rises from 0 at disk centre to 11.7 % at the limb (data: Chandrasekhar 1950, Table XXIV). Inset: the limb darkening differs for the two polarization states.

Quantity	Definition
degree of polarization	$p = \sqrt{Q^2 + U^2 + V^2}\,/\,I \leq 1$
degree of linear polarization	$p_L = \sqrt{Q^2 + U^2}\,/\,I$
polarization angle	$\chi = \frac{1}{2}\arctan(U/Q)$
circular polarization	$p_V = V/I$
fully polarized light	$I^2 = Q^2 + U^2 + V^2$

Rotation of axes by an angle ϕ: $Q' = Q\cos 2\phi + U \sin 2\phi$, $U' = -Q \sin 2\phi + U \cos 2\phi$, with I and V unchanged. The factor 2ϕ means linear polarization is a "spin-2" quantity, the basis of the E- and B-mode decomposition of CMB polarization.

5.2 The polarized transfer equation

Write the Stokes vector $\mathbf{I} = (I, Q, U, V)^T$. The transfer equation becomes a matrix equation:

$$\boxed{\frac{d\mathbf{I}_\nu}{ds} = -\mathsf{K}_\nu\,\mathbf{I}_\nu + \boldsymbol{\epsilon}_\nu} \qquad \mathsf{K} = \begin{pmatrix} \eta_I & \eta_Q & \eta_U & \eta_V \\ \eta_Q & \eta_I & \rho_V & -\rho_U \\ \eta_U & -\rho_V & \eta_I & \rho_Q \\ \eta_V & \rho_U & -\rho_Q & \eta_I \end{pmatrix}$$

- η_I: ordinary absorption. $\eta_{Q,U,V}$: **dichroism**, meaning different absorption for different polarization states.
- $\rho_{Q,U,V}$: **magneto-optical (Faraday) effects**, rotating the polarization plane and converting linear into circular polarization.
- $\boldsymbol{\epsilon}$: emission vector. In LTE $\boldsymbol{\epsilon} = \mathsf{K}\,(B_\nu, 0, 0, 0)^T$, the polarized form of Kirchhoff's law.
- For an unmagnetized, isotropic medium $\mathsf{K} = \alpha_\nu \mathbf{1}$, and the equations for I, Q, U, V decouple.
- **Scattering** adds a gain term with the **phase matrix** $\mathsf{P}(\hat{\mathbf{n}}', \hat{\mathbf{n}})$ (Chandrasekhar's Rayleigh phase matrix for Thomson and Rayleigh scattering).
- With Zeeman-split absorption profiles these are the **Unno–Rachkovsky equations** (Unno 1956; Rachkovsky 1962). Sign conventions differ between authors (see Landi Degl'Innocenti & Landolfi 2004).

5.3 Polarization by scattering

For Thomson or Rayleigh scattering of unpolarized light through a scattering angle Θ:

$$I_{\rm sc} \propto \frac{1 + \cos^2\Theta}{2}, \qquad p_L = \frac{1 - \cos^2\Theta}{1 + \cos^2\Theta} = \frac{\sin^2\Theta}{1 + \cos^2\Theta}.$$

So the light is unpolarized in the forward and backward directions and 100 % polarized at $\Theta = 90^\circ$.

Chandrasekhar's electron-scattering atmosphere (1946). In a semi-infinite atmosphere where Thomson scattering dominates (hot O and B stars), the radiation emerging near the limb has travelled mostly parallel to the surface before its last scattering. It is therefore polarized, with the electric vector **parallel to the limb**. The exact solution of the polarized transfer equation gives:

μ	0 (limb)	0.1	0.2	0.4	0.6	0.8	1 (centre)
degree of polarization	11.71 %	7.45 %	5.41 %	3.03 %	1.63 %	0.68 %	0

For an unresolved spherical star the polarization cancels by symmetry. It becomes observable when the symmetry is broken: in eclipsing binaries (where it is measured during eclipses), rapidly rotating oblate stars, or disks.

5.4 Magnetic fields: the Zeeman and Hanle effects

Zeeman splitting. A field B splits a line by

$$\Delta\lambda_B = 4.67 \times 10^{-13}\, g_{\text{eff}}\, \lambda^2\, B,$$

with λ and $\Delta\lambda_B$ in ångström (Å) and B in gauss (G), where g_{eff} is the effective Landé factor. The σ components are circularly polarized when viewed along the field and linearly polarized when viewed across it; the π component is linearly polarized.

Weak-field approximation (splitting much smaller than the Doppler width): $V(\lambda) \approx -\Delta\lambda_B \cos\gamma \, \dfrac{\partial I}{\partial \lambda}$, where γ is the angle between field and line of sight. V is antisymmetric about line centre and proportional to the longitudinal field, which is the principle of **magnetographs** (e.g. SDO/HMI).

Hanle effect. Weak fields (roughly 1–100 G, depending on the line) change the polarization of *scattered* line radiation (depolarization and rotation). The linearly polarized spectrum of the solar limb, the “second solar spectrum” (Stenflo & Keller 1997), is a diagnostic of turbulent magnetic fields that the Zeeman effect cannot detect.

5.5 Where polarized transfer is used

Field	Mechanism	Example
solar physics	Zeeman, Hanle	sunspot and quiet-Sun magnetic maps; Milne–Eddington inversions of Stokes profiles
hot stars, binaries, disks	Thomson scattering	eclipse polarization; Be-star disk polarization
interstellar medium	dichroic extinction by aligned dust (Hiltner and Hall, 1949)	Galactic magnetic-field direction
radio galaxies, pulsars, blazars	synchrotron emission	up to ~70 % linear polarization; Faraday rotation measures
cosmology	Thomson scattering of a quadrupole anisotropy	CMB E-modes (first detected by DASI in 2002); B-modes
planets and exoplanets	Rayleigh scattering	sky polarization; reflected light of planets

Worked example 5.1 — Stokes parameters from filter measurements

Through six filters a detector measures (in units of the total): horizontal 0.70, vertical 0.30, +45° 0.60, −45° 0.40, right-circular 0.50, left-circular 0.50. Find I, Q, U, V, p and χ.

$I = 0.70 + 0.30 = 1.00$, $Q = 0.40$, $U = 0.20$, $V = 0$. So $p = \sqrt{0.40^2 + 0.20^2} = 0.447$ (44.7 % linearly polarized) and $\chi = \tfrac{1}{2}\arctan(0.20/0.40) = 13.3^\circ$.

Worked example 5.2 — Polarization of scattered light

Unpolarized sunlight is Thomson-scattered by electrons. Find the degree of polarization at $\Theta = 90^\circ$, 60° and 30°.

$p = \sin^2\Theta/(1+\cos^2\Theta)$: 90° → $1/1 = 100\,\%$; 60° → $0.75/1.25 = 60\,\%$; 30° → $0.25/1.75 = 14\,\%$. The same law gives the polarization pattern of the blue sky (Rayleigh scattering) and of the solar corona (the K-corona, Thomson scattering by free electrons).

Worked example 5.3 — Is a line split in a sunspot?

The Fe I line at 6302.5 Å has $g_{\rm eff} = 2.5$. Compute the Zeeman splitting for $B = 3000$ G (sunspot umbra) and $B = 100$ G (plage), and compare with the Doppler width (about 28 mÅ at 5800 K, Lecture SA2).

$\Delta\lambda_B = 4.67\times10^{-13}\times 2.5 \times (6302.5)^2 \times B = 4.64\times10^{-5}\,B$ Å.

- 3000 G: $\Delta\lambda_B = 0.139$ Å $= 139$ mÅ, 5 times the Doppler width. The line is visibly split (strong-field regime).
- 100 G: $\Delta\lambda_B = 4.6$ mÅ, much smaller than the Doppler width. The line is not split, but $V \propto \partial I/\partial\lambda$ still measures the field (weak-field regime).

Common misconceptions

1. **“Thermal emission is polarized.”** Isotropic thermal emission in LTE is unpolarized; polarization requires an asymmetry (scattering geometry, magnetic field, aligned grains).
2. **“Q and U are properties of the source alone.”** They depend on the chosen reference axes; only p and, after fixing axes, χ have direct physical meaning.
3. **“A star’s limb polarization is easy to see.”** For an unresolved round star it averages to zero; it is measured only when the symmetry is broken (eclipses, disks, rotation).
4. **“The Zeeman effect only works for strong fields.”** Even when the splitting is much smaller than the line width, circular polarization measures the longitudinal field (weak-field approximation).

Lecture summary

- Stokes vector (I, Q, U, V): total, two linear polarizations (0°/90° and ±45°), circular.
- Polarized RTE: $d\mathbf{I}/ds = -\mathbf{K}\mathbf{I} + \boldsymbol{\epsilon}$, with absorption, dichroism and magneto-optical terms; the Unno–Rachkovsky equations.
- Thomson and Rayleigh scattering: $p = \sin^2\Theta/(1+\cos^2\Theta)$; an electron-scattering atmosphere has 11.7 % polarization at the limb (Chandrasekhar).
- Zeeman: $\Delta\lambda_B = 4.67\times10^{-13} g \lambda^2 B$; V measures the longitudinal field. Hanle effect: weak fields.

Exam questions and answers — Lecture RT5

Q1. Define the four Stokes parameters operationally. Show that for completely polarized light $I^2 = Q^2 + U^2 + V^2$.

> **A1.** $I = I_1 + I_2$, $Q = I_1 - I_2$ (0°/90° polarizers), $U = I_{1'} - I_{2'}$ (±45°), $V = I_+ - I_-$ (circular). For a fully polarized wave, $I = a_1^2 + a_2^2$, $Q = a_1^2 - a_2^2$, $U = 2a_1 a_2 \cos\delta$, $V = 2a_1 a_2 \sin\delta$, so $Q^2 + U^2 + V^2 = (a_1^2 - a_2^2)^2 + 4a_1^2 a_2^2 = (a_1^2 + a_2^2)^2 = I^2$.

Q2. Why is light scattered by an electron at 90° completely polarized? What is the degree of polarization at 45°?

> **A2.** The electron oscillates along the incident electric field, which is perpendicular to the incoming direction. A dipole does not radiate along its axis, so at 90° only the component of oscillation perpendicular to the scattering plane is seen: 100 % linear polarization. At 45°: $p = \sin^2 45^\circ / (1 + \cos^2 45^\circ) = 0.5/1.5 = 33\,\%$.

Q3. Describe Chandrasekhar's result for the polarization of an electron-scattering atmosphere. Why is it hard to observe, and how can it be detected?

> **A3.** In a semi-infinite pure Thomson-scattering atmosphere, the emergent polarization rises from 0 at disk centre to 11.7 % at the limb, with the electric vector parallel to the limb. For a spherical, unresolved star the contributions from opposite limbs cancel. It becomes observable when symmetry is broken: during eclipses of binaries, in rotationally flattened stars, or in circumstellar disks.

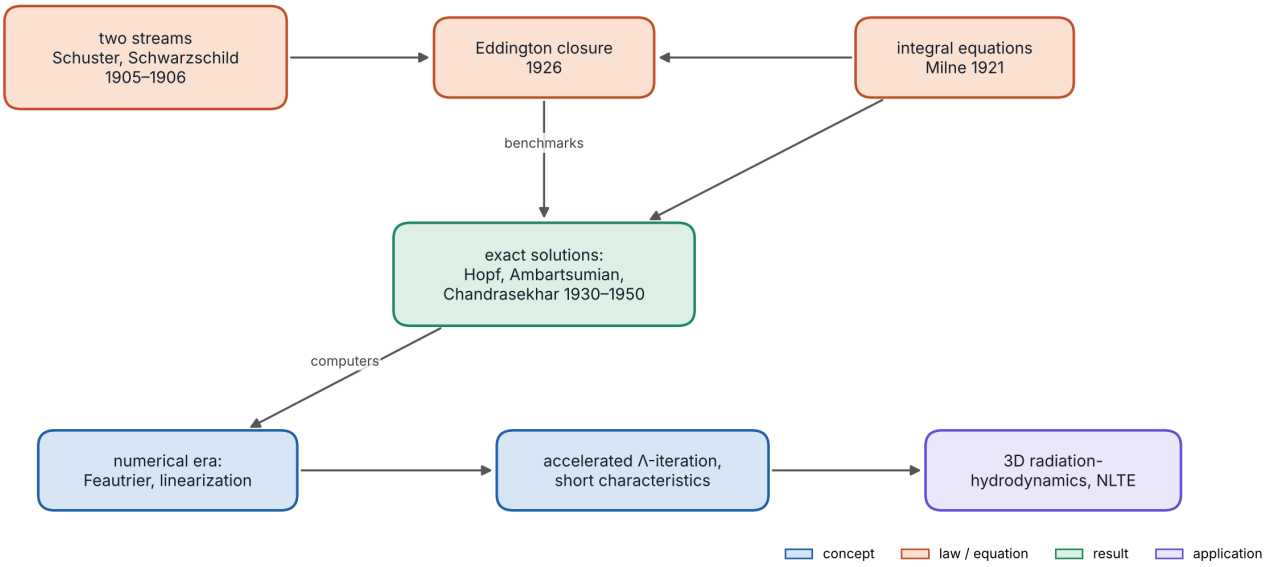
Q4. What is the Zeeman splitting of a line with $g = 1.5$ at 5000 Å in a 1000 G field? In which regime is it, and how would you measure the field?

> **A4.** $\Delta\lambda_B = 4.67 \times 10^{-13} \times 1.5 \times (5000)^2 \times 1000 = 0.0175$ Å $= 17.5$ mÅ. This is smaller than a typical Doppler width (about 20–30 mÅ), so it is in the weak-field regime and the line is not visibly split. Measure the circular polarization: $V \approx -\Delta\lambda_B \cos\gamma \, \partial I/\partial\lambda$ gives the longitudinal component $B\cos\gamma$.

Lecture RT6 — How the Problem Was Solved: a History of Approximations and Solutions

> **Sources:** Chandrasekhar, *Radiative Transfer* (1950), Preface and bibliographical notes of Chs. II–V (Schuster, Schwarzschild, Milne, Eddington, Hopf, Bronstein, Wiener–Hopf, Ambartsumian); Collins Ch. 9 and its supplemental reading (Eddington 1916; Mihalas 1978; Sobolev 1963; Mustel 1958); A670 Lecture RT1 (historical background); Hubeny & Mihalas, *Theory of Stellar Atmospheres* (2014), for the numerical era.

RT6 concept map — how the problem was solved



*Concept map — the story in three acts. **Approximations** (two streams, Eddington's closure, Milne's integral equations) came first, because they could be done by hand. **Exact solutions** (Hopf, Ambartsumian, Chandrasekhar) followed and became benchmarks. **Numerical methods** took over from the 1960s and now run 3D radiation-hydrodynamics.*

Learning goals

1. Place the main approximations of this course (two-stream, Eddington, diffusion, Milne–Eddington, Eddington–Barbier) in their historical setting.
2. Know who found the exact solutions of the classical problems, and by which methods.
3. Name the main numerical methods of the computer era and the problem each one solved.
4. Compare the accuracy of the historical approximations on the gray atmosphere.

The big ideas in plain words

Why approximations came first. Until the 1960s there were no computers able to solve integro-differential equations. Astronomers needed answers they could compute with pencil, paper and tables. They invented approximations that replace the full angular dependence of the radiation by one or two numbers (two streams, or the Eddington closure

$K = J/3$). Remarkably, these simple ideas already explained limb darkening, the temperature structure of the Sun and the formation of absorption lines.

Exact solutions as benchmarks. Between 1930 and 1950, mathematicians and astrophysicists (Hopf, Wiener, Ambartsumian, Chandrasekhar) solved a few idealised problems *exactly*: the gray atmosphere, isotropic scattering, the polarization of scattered light. These solutions, such as the Hopf function and the H-functions, are still used to test every new numerical code, exactly as we did with our own solver in Lecture SA4.

The numerical era: solving the hard problems. Real atmospheres need thousands of frequencies, many atomic levels and NLTE populations that depend on the radiation field. From the 1960s onwards, new algorithms turned this into problems computers could solve: Feautrier's method, complete linearization, accelerated Λ-iteration, short characteristics, Monte Carlo. Today's 3D radiation-hydrodynamic simulations build on all of them.

Old approximations never die. Two-stream and Eddington approximations are still used in climate models, exoplanet atmospheres and supernova codes. The Milne–Eddington model is the basis of solar magnetic-field measurements. Understanding the history explains why these "old" formulas appear in modern papers.

6.1 Before transfer theory (1729–1917)

Year	Who	Contribution
1729, 1760, 1852	Bouguer, Lambert, Beer	exponential attenuation of light, $I = I_0 e^{-\tau}$
1859–1860	Kirchhoff (with Bunsen)	emission/absorption ratio is universal; spectral analysis; dark solar lines explained
1900	Planck	the universal function $B_\nu(T)$
1917	Einstein	A and B coefficients; stimulated emission
1920	Saha	ionisation equation, which made the spectral sequence a temperature sequence

6.2 The birth of stellar radiative transfer (1905–1930)

- **Schuster (1905)**, "Radiation through a foggy atmosphere": the first transfer problem with scattering, treated with **two streams** (one up, one down). He explained how a scattering layer can produce absorption or emission lines.
- **Karl Schwarzschild (1906)**: showed that the solar photosphere is in **radiative equilibrium**, not convective (adiabatic) equilibrium. With a two-stream approximation he found $T^4 = \frac{1}{2}T_{\rm eff}^4(1 + 2\tau)$ and the limb-darkening law $I(\mu)/I(1) = (1 + 2\mu)/3$, close to the observed solar limb darkening. In 1914 he wrote the theory of scattering lines.
- **Rosseland (1924)**: the diffusion limit and the **Rosseland mean** opacity (Lecture RT4).
- **Milne (1921)**: formulated the transfer problem as **integral equations** (the Schwarzschild–Milne equations and the "Milne problem" of Lecture SA4), and with Eddington the line-formation model now called **Milne–Eddington** (Lecture SA4).

- **Eddington (1916, 1926)**: radiative equilibrium of whole stars; the **Eddington approximation** $K = J/3$ with $J(0) = 2H(0)$ (Lecture RT4), which gave $T^4 = \frac{3}{4}T_{\rm eff}^4(\tau + \frac{2}{3})$.
- **Unsöld (1928–1938)**: quantitative abundance analysis; the solar atmosphere is mostly hydrogen. His book *Physik der Sternatmosphären* (1938) became the standard text.

6.3 The era of exact solutions (1930–1960)

- **Hopf (1930–1934)** and **Bronstein (1929)**: exact solution of the Milne problem; Hopf–Bronstein relation $J(0) = \sqrt{3}H$; Hopf function $q(\tau)$ with $q(\infty) = 0.7104$.
- **Wiener & Hopf (1931)**: the Wiener–Hopf technique for integral equations on a half-line, first motivated by this problem.
- **Wildt (1939)**: the negative hydrogen ion H^- is the main continuum opacity of the Sun. Chandrasekhar then computed its cross-sections (1940s).
- **Barbier (1943)**: the relation $I(0, \mu) \approx S(\tau = \mu)$, now the **Eddington–Barbier relation** (Lecture RT4).
- **Ambartsumian (1943)**: **principles of invariance**. Adding a thin layer to a semi-infinite atmosphere cannot change its reflected light, which gives equations for the emergent radiation without solving for the interior.
- **Chandrasekhar (1944–1950)**: the **method of discrete ordinates** (Gaussian quadrature in angle), the H-**functions**, the exact law of darkening, the polarization of electron-scattering atmospheres (11.7 % at the limb, 1946) and the sunlit-sky (Rayleigh) problem. Collected in *Radiative Transfer* (1950).
- **Kourganoff (1952)**, *Basic Methods in Transfer Problems*; **Sobolev (1947, 1957, 1963)**: the **Sobolev approximation** for rapidly expanding envelopes and his *Treatise on Radiative Transfer*.
- **R. N. Thomas (1957–1965)**: NLTE line source functions, the two-level atom, thermalisation length (Lectures RT4 and SA3).

6.4 The numerical era (1960–today)

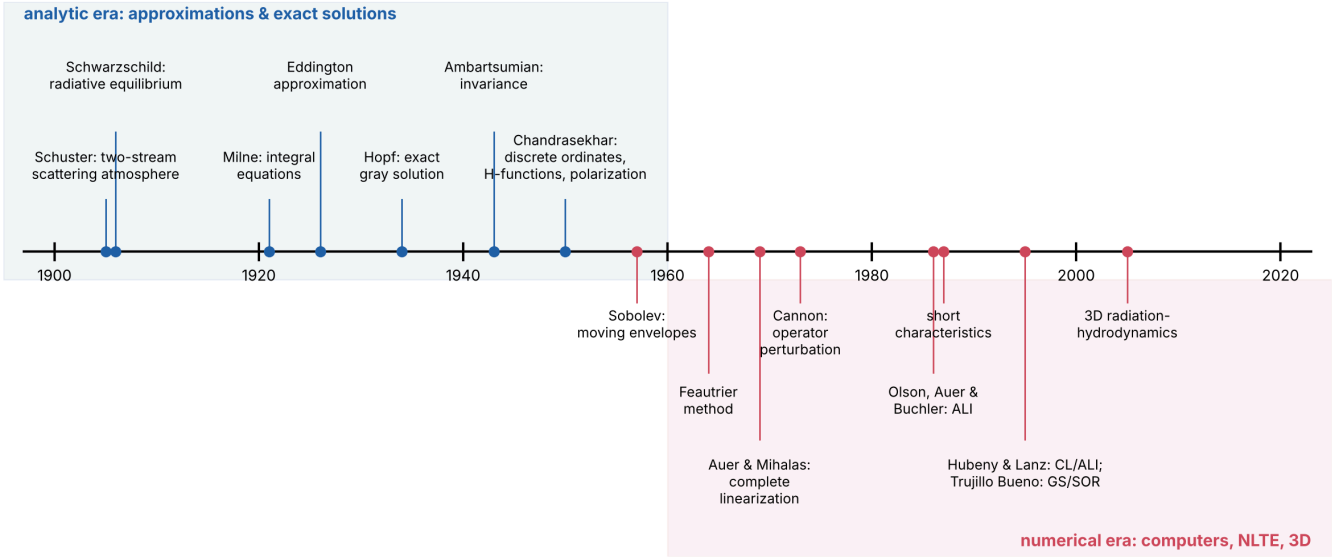


Figure 6.1 — Two eras. Above the axis: approximations and exact solutions. Below: numerical methods. Dates are approximate and refer to the first major publication.

Year	Method	Problem it solved
1964	**Feautrier** method (second-order difference equations)	stable solution of the RTE with scattering
1969	**Complete linearization** (Auer & Mihalas)	first self-consistent NLTE model atmospheres
1970	**Variable Eddington factors** (Auer & Mihalas)	closes the moment equations accurately
1970	**ATLAS** code (Kurucz)	LTE line-blanketed model atmosphere grids
1971–1973	**Rybicki method**; **operator perturbation** (Cannon)	efficient scattering problems; seed of ALI
1975	**MARCS** (Gustafsson et al.)	models of cool stars with molecules
1981–1986	**Accelerated Λ-iteration** (Scharmer; Werner & Husfeld; Olson, Auer & Buchler)	fast NLTE solutions with an approximate Λ operator
1987	**Short characteristics** (Olson & Kunasz)	multi-dimensional formal solutions
1991	**MALI** (Rybicki & Hummer)	multi-level NLTE atoms
1995	**Hybrid CL/ALI** (Hubeny & Lanz); **Gauss–Seidel/SOR** (Trujillo Bueno & Fabiani Bendicho)	NLTE metal-line-blanketed models; faster convergence
1982–today	**3D radiation-hydrodynamics** (Nordlund; Stein & Nordlund; Asplund et al.)	convection and granulation; revised solar abundances
1960s–today	**Monte Carlo** radiative transfer	complex geometry: dust, disks, winds, polarization

6.5 How good were the historical approximations?

Applied to the same gray atmosphere:

Approximation	$T^4/T_{\rm eff}^4$	$T(0)/T_{\rm eff}$	limb darkening $I(\mu)/I(1)$	$I(0)/I(1)$
Schwarzschild (two-stream, $\mu = \pm\frac{1}{2}$)	$\tau + \frac{1}{2}$	0.841	$(1+2\mu)/3$	0.333
Two-stream with $\mu = \pm 1/\sqrt{3}$	$\frac{3}{4}(\tau + 0.577)$	0.811	$(1+\sqrt{3}\mu)/(1+\sqrt{3})$	0.366
Eddington	$\frac{3}{4}(\tau + \frac{2}{3})$	0.841	$(2+3\mu)/5$	0.400
Exact (Hopf, Chandrasekhar)	$\frac{3}{4}(\tau + q(\tau))$	**0.811**	$H(\mu)/H(1)$	**0.344**

All the approximations are within about 15 % of the exact answer, accurate enough to have guided astrophysics for half a century.

Worked example 6.1 — Reproducing Schwarzschild's 1906 result

Use two streams at $\mu = \pm\frac{1}{2}$ (I^+ outward, I^- inward) in a gray LTE atmosphere in radiative equilibrium.

With these streams: $J = \frac{1}{2}(I^+ + I^-)$, $H = \frac{1}{4}(I^+ - I^-)$, $K = \frac{1}{8}(I^+ + I^-) = J/4$. Radiative equilibrium gives H = const and $dK/d\tau = H$, so $J = 4H\tau + C$. At the surface $I^- = 0$: $J(0) = I^+/2$ and $H(0) = I^+/4$, so $J(0) = 2H$ and $C = 2H$. Then $J = 4H(\tau + \frac{1}{2})$, and with $J = \sigma T^4/\pi$, $H = \sigma T_{\rm eff}^4/4\pi$:

$$T^4 = T_{\rm eff}^4 \left(\tau + \tfrac{1}{2}\right) = \tfrac{1}{2} T_{\rm eff}^4 (1 + 2\tau).$$

The emergent intensity, by Eddington–Barbier (exact for linear S), is $\propto 1 + 2\mu$, which is Schwarzschild's limb-darkening law.

Worked example 6.2 — Which approximation is closest at the limb?

Using the table, compute the error of $I(0)/I(1)$ for each approximation.

Schwarzschild: $(0.333 - 0.344)/0.344 = -3.2\,\%$. Two-stream with $\mu = 1/\sqrt{3}$: $+6.4\,\%$. Eddington: $+16\,\%$. The oldest approximation is the most accurate at the limb, although Eddington's is better on average and for the temperature deep in the atmosphere ($q(\infty) = 0.710$ against Eddington's 0.667).

Common misconceptions

1. **"The Eddington approximation is obsolete."** Variable-Eddington-factor methods, climate models and many astrophysical codes still use it as a closure.
2. **"Exact solutions exist for realistic atmospheres."** They exist only for idealised problems (gray, isotropic scattering); realistic problems are numerical.
3. **"Numerical methods replaced the theory."** Numerical codes are tested against the exact solutions (Hopf, H-functions), as our own solver was.
4. **"Radiative transfer started with computers."** The key physical ideas (radiative equilibrium, limb darkening, line formation, scattering) were all established before 1930.

Lecture summary

- 1905–1930: Schuster and Schwarzschild (two streams), Milne (integral equations), Eddington ($K = J/3$), Rosseland (diffusion).
- 1930–1960: exact solutions by Hopf, Wiener–Hopf, Ambartsumian and Chandrasekhar; Sobolev's approximation for moving media.
- 1960–today: Feautrier, complete linearization, ALI/MALI, short characteristics, Monte Carlo, 3D radiation-hydrodynamics.
- The historical approximations are within about 15 % of the exact gray solution, and they are still used today.

Exam questions and answers — Lecture RT6

Q1. What did Karl Schwarzschild show in 1906, and why was it important?

> **A1.** He showed that the solar photosphere is in radiative, not adiabatic (convective), equilibrium. Using a two-stream approximation he derived $T^4 = \frac{1}{2}T_{\rm eff}^4(1+2\tau)$ and the limb-darkening law $(1+2\mu)/3$, which matched solar observations. This established radiative equilibrium and the transfer equation as the basis of stellar-atmosphere theory.

Q2. Explain the difference between the two-stream approximation and the Eddington approximation.

> **A2.** The two-stream approximation replaces the full angular distribution by two beams at fixed angles $\mu = \pm\mu_0$, solving two coupled ordinary differential equations. The Eddington approximation keeps the full angular range but closes the moment equations with $K = J/3$ (exact for a field linear in μ) and uses the surface condition $J(0) = 2H(0)$. For $\mu_0 = 1/\sqrt{3}$ the two-stream method also has $K = J/3$, but its surface condition is $J(0) = \sqrt{3}H$.

Q3. Name three contributions of Chandrasekhar to radiative transfer.

> **A3.** (1) The method of discrete ordinates (Gaussian quadrature in angle) with explicit solutions in every order. (2) The H-functions and the exact laws of darkening for isotropic and Rayleigh scattering. (3) The polarized transfer equation and the prediction of 11.7 % limb polarization in electron-scattering atmospheres; he also solved the sunlit-sky (Rayleigh scattering) problem.

Q4. Which numerical method first made self-consistent NLTE model atmospheres possible, and which later family of methods made them fast?

> **A4.** Complete linearization (Auer & Mihalas 1969) first produced self-consistent NLTE models. Accelerated Λ-iteration (operator perturbation: Cannon 1973; Scharmer 1981; Olson, Auer & Buchler 1986), its multi-level form MALI (Rybicki & Hummer 1991) and the hybrid CL/ALI scheme (Hubeny & Lanz 1995) made them fast enough for many atomic levels and lines.

Final Examination — Course A: Radiative Transfer (with complete solutions)

Format suggestion: 3 hours. Part A (20 %): short answers. Part B (50 %): answer any 6 of the 8 problems. Part C (20 %): RT3, RT5 and RT6 topics. Part E (10 %): essay. Solutions follow each question; numbers were checked with `scripts/worked_examples.py` and `scripts/lte_solvers.py`. Constants: $h = 6.626 \times 10^{-34}$ J s, $k = 1.381 \times 10^{-23}$ J K^{-1}, $c = 2.998 \times 10^{8}$ m s^{-1}, $\sigma = 5.670 \times 10^{-8}$ W m^{-2} K^{-4}, $m_e = 9.109 \times 10^{-31}$ kg, $m_H = 1.674 \times 10^{-27}$ kg, $r_e = 2.818 \times 10^{-15}$ m, $\sigma_T = 6.652 \times 10^{-29}$ m^2, $R_\odot = 6.957 \times 10^{8}$ m, 1 pc $= 3.086 \times 10^{16}$ m, 1 eV $= 1.602 \times 10^{-19}$ J. $\int_0^\infty E_n(t)\,dt = \dfrac{1}{n}$, $\int_0^\infty t\,E_n(t)\,dt = \dfrac{1}{n+1}$, $E_n(0) = \dfrac{1}{n-1}$.

Part A — Short answers (2 marks each)

A1. What is conserved along a ray in vacuum: F_ν, I_ν or J_ν?

> I_ν (more generally I_ν/ν^3).

A2. Give the source function for coherent isotropic scattering plus thermal absorption.

> $S_\nu = \varepsilon B_\nu + (1-\varepsilon) J_\nu$, with $\varepsilon = \kappa^a/(\kappa^a + \sigma^{\rm sc})$.

A3. At what optical depth does the emergent flux “originate” according to Eddington–Barbier?

> $\tau_\nu = 2/3$: $F_\nu(0) \approx \pi S_\nu(2/3)$.

A4. Why is the Rosseland mean a harmonic mean?

> The flux in each frequency band is $\propto 1/\kappa_\nu$ (like conductances in parallel), so the total is dominated by low-opacity windows.

A5. Write the Eddington closure and the Eddington surface condition.

> $K = J/3$; $J(0) = 2H(0)$.

A6. What does the factor $1 - e^{-h\nu/kT}$ represent?

> The LTE correction for stimulated emission (negative absorption).

A7. For an isothermal scattering atmosphere, what is $S(0)/B$ exactly?

> $\sqrt{\varepsilon}$.

A8. What is an optically thin source's brightness temperature in terms of T and τ?

> $T_b = T(1 - e^{-\tau}) \approx T\tau$ (RJ, no background).

A9. Two stars with equal $T_{\rm eff}$ but different radii: which quantity is the same, the flux at Earth or the surface brightness?

> The surface brightness.

Part B — Problems

B1. Intensity and flux (8 marks)

An A-type star has $R = 1.7 R_\odot$ and $T_{\rm eff} = 9900$ K and lies at $d = 2.64$ pc. (a) Find its surface flux and luminosity in $L_\odot$ ($L_\odot = 3.828 \times 10^{26}$ W). (b) Find the bolometric flux at Earth. (c) Show that its mean surface brightness is the same at the star and at Earth.

> **Solution.** (a) $F_s = \sigma T^4 = 5.67 \times 10^{-8} \times (9900)^4 = 5.45 \times 10^{8}$ W m^{-2}; $L = 4\pi R^2 F_s = 25\, L_\odot$. (b) $F_\oplus = F_s (R/d)^2$, with $R = 1.18 \times 10^{9}$ m and $d = 8.15 \times 10^{16}$ m, so $(R/d)^2 = 2.1 \times 10^{-16}$ and $F_\oplus = 1.15 \times 10^{-7}$ W m^{-2}. (c) The mean intensity is $\bar I = F_s/\pi$. At Earth, $\bar I = F_\oplus / \Omega$ with $\Omega = \pi (R/d)^2$, giving $F_s (R/d)^2 / (\pi (R/d)^2) = F_s/\pi$, the same. Intensity is conserved along rays.

B2. Planck function (8 marks)

a. Show that in the RJ limit $I_\nu = (2k\nu^2/c^2) T_b$. (b) A compact source has $I_\nu = 10^{-18}$ W m^{-2} Hz^{-1} sr^{-1} at 5 GHz. Find T_b. (c) If the source is an H II region with $T_e = 10^4$ K, what is its optical depth at 5 GHz? (d) Is it thick or thin at 0.5 GHz if $\tau \propto \nu^{-2.1}$?

> **Solution.** (a) $e^x - 1 \approx x$ gives $B_\nu \approx 2h\nu^3/(c^2 x) = 2\nu^2 kT/c^2$; define T_b by $I_\nu = B_\nu(T_b)$. (b) $T_b = c^2 I_\nu / (2k\nu^2) = (8.99 \times 10^{16} \times 10^{-18}) / (2 \times 1.381 \times 10^{-23} \times 2.5 \times 10^{19}) = 130$ K. (c) $T_b = T_e(1 - e^{-\tau}) \Rightarrow 1 - e^{-\tau} = 0.013 \Rightarrow \tau = 0.013$. (d) $\tau(0.5\ \text{GHz}) = 0.013 \times 10^{2.1} = 1.6$: marginally thick, so $T_b = 10^4 (1 - e^{-1.6}) = 8000$ K.

B3. Formal solution in a plane-parallel atmosphere (8 marks)

$S(\tau) = a + b\tau + c\tau^2$ in a semi-infinite atmosphere. (a) Find $I(0,\mu)$. (b) Find $F(0)$. (c) Compare with the Eddington–Barbier values and state when they are exact.

> **Solution.** (a) $I(0,\mu) = \int_0^\infty (a + bt + ct^2) e^{-t/\mu} dt/\mu = a + b\mu + 2c\mu^2$. (b) $F = 2\pi \int_0^1 (a + b\mu + 2c\mu^2)\mu\, d\mu = 2\pi(a/2 + b/3 + c/2) = \pi(a + \tfrac{2}{3}b + c)$. (c) E–B: $I \approx S(\mu) = a + b\mu + c\mu^2$ (error $c\mu^2$); $F \approx \pi S(2/3) = \pi(a + \tfrac{2}{3}b + \tfrac{4}{9}c)$ (error $\tfrac{5}{9}\pi c$). Both are exact when $c = 0$ (linear S).

B4. Testing the Eddington solution with the Λ operator (8 marks)

For the Eddington gray solution $S(\tau) = 3H(\tau + \tfrac{2}{3})$, compute $J(0) = \Lambda_0[S] = \tfrac{1}{2}\int_0^\infty S(t) E_1(t)\, dt$. Compare with $S(0)$ and with the exact (Hopf–Bronstein) value $J(0) = \sqrt{3}H$. What does this tell you?

> **Solution.** $J(0) = \tfrac{3H}{2}\left[\int_0^\infty t E_1 dt + \tfrac{2}{3}\int_0^\infty E_1 dt\right] = \tfrac{3H}{2}\left[\tfrac{1}{2} + \tfrac{2}{3}\right] = \tfrac{7}{4}H = 1.750\,H$. But $S(0) = 2H$, so $\Lambda[S] \neq S$: the Eddington solution does **not** satisfy Milne's equation exactly ($J \neq S$ at the surface). The exact value $\sqrt{3}H = 1.732H$ is close to the result of one Λ-iteration (1.750H), which shows that Λ-iteration improves the solution near the surface, where it couples to the boundary.

B5. Two-stream (Schuster–Schwarzschild) gray atmosphere (10 marks)

Represent the radiation field by two streams, I^+ at $\mu = +1/\sqrt{3}$ and I^- at $\mu = -1/\sqrt{3}$, so that $J = \tfrac{1}{2}(I^+ + I^-)$, $H = \tfrac{1}{2\sqrt{3}}(I^+ - I^-)$, $K = \tfrac{1}{6}(I^+ + I^-)$. (a) Show that $K = J/3$ exactly. (b) With $I^-(0) = 0$, show $J(0) = \sqrt{3}H$. (c) Hence derive $T^4 = \tfrac{3}{4}T_{\rm eff}^4(\tau + 1/\sqrt{3})$ and compare $T(0)$ with the exact result.

> **Solution.** (a) $K = \tfrac{1}{2}\sum \mu_i^2 I_i = \tfrac{1}{2}\cdot\tfrac{1}{3}(I^+ + I^-) = J/3$. (b) $J(0) = I^+/2$ and $H(0) = I^+/(2\sqrt{3})$, so $J(0) = \sqrt{3}H(0)$. (c) As in B6: $J = 3H(\tau + C)$ with $3HC = \sqrt{3}H \Rightarrow C = 1/\sqrt{3}$. So $T^4 = \tfrac{3}{4}T_{\rm eff}^4(\tau + 0.577)$ and $T(0) = (\sqrt{3}/4)^{1/4} T_{\rm eff} = 0.811\,T_{\rm eff}$, which is **exactly the true surface value**. This is Chandrasekhar's "first approximation" of the discrete-ordinate method; it gives $J(0)$ exactly in every order (Chandrasekhar eq. 64), while $q(\infty)$ improves with the order.

B6. Limb darkening and LTE temperature structure (8 marks)

At $\lambda = 500$ nm, the observed solar limb darkening is $I(\mu) = I(1)(0.36 + 0.64\mu)$ with $I(1) = 4.0 \times 10^{13}$ W m^{-3} sr^{-1}. Assuming LTE and the Eddington–Barbier relation, find $T(\tau_{500} = 0)$ and $T(\tau_{500} = 1)$. Comment.

> **Solution.** $S(\tau) = I(1)(0.36 + 0.64\tau)$, and LTE gives $S = B_\lambda(T)$, so $T = (hc/\lambda k)/\ln(1 + 2hc^2/(\lambda^5 B))$, with $hc/\lambda k = 28\,776$ K and $2hc^2/\lambda^5 = 3.811 \times 10^{15}$. $\tau = 0$: $B = 1.44 \times 10^{13}$, so $T = 28776/\ln(265.7) = 5155$ K. $\tau = 1$: $B = 4.0 \times 10^{13}$, so $T = 28776/\ln(96.3) = 6300$ K. The temperature rises inward by about 1100 K per unit optical depth, which is why the Sun shows limb darkening and absorption lines. The values bracket $T_{\rm eff} = 5772$ K, consistent with the gray model (Lecture SA4).

B7. Scattering and thermalisation (10 marks)

An isothermal atmosphere has $S = \varepsilon B + (1-\varepsilon) J$. (a) In the Eddington approximation, derive the differential equation for J and solve it with the appropriate boundary conditions. (b) For $\varepsilon = 10^{-2}$, find $S(0)/B$ (Eddington) and compare with the exact value. (c) Estimate the thermalisation depth and the number of scatterings a photon undergoes before destruction. (d) Why does Λ-iteration starting from $S = B$ converge so slowly?

> **Solution.** (a) $dH/d\tau = J - S = \varepsilon(J - B)$ and $H = \tfrac{1}{3} dJ/d\tau$, so $\tfrac{1}{3} J'' = \varepsilon (J - B)$. Then $J = B + C e^{-\sqrt{3\varepsilon}\tau}$; with $J(0) = \tfrac{2}{3} J'(0)$, $C = -B/(1 + 2\sqrt{\varepsilon/3})$. (b) $\sqrt{\varepsilon/3} = 0.0577$, so $J(0)/B = 1 - 1/1.1155 = 0.1035$ and $S(0)/B = 0.01 + 0.99 \times 0.1035 = 0.1125$. The exact value is $\sqrt{\varepsilon} = 0.100$; Eddington overestimates by 12 %. (c) $\tau_{\rm th} \sim 1/\sqrt{\varepsilon} = 10$ (Eddington: $1/\sqrt{3\varepsilon} = 5.8$); $N \sim 1/\varepsilon = 100$ scatterings. (d) Each iteration transmits information only about $\Delta\tau \sim 1$, and the error decreases by a factor $(1-\varepsilon) = 0.99$ per step, so hundreds of iterations are needed (see Lecture RT4, Figure 4.4). Our solver gives $S(0)$ errors of 95 % after 10 iterations, 17 % after 50, 5 % after 100 and 0.9 % after 200; yet beyond 100 iterations each step changes $S(0)$ by less than 0.1 %, so a naive convergence test would stop far too early.

B8. Diffusion and Rosseland mean (8 marks)

a. From $I \approx B + \mu\, dB/d\tau$, derive $F = \frac{4\pi}{3} \frac{dB}{d\tau}$. (b) A gas has three frequency bands carrying 50 %, 30 % and 20 % of $\partial B/\partial T$, with $\kappa = 1$, 10 and 1000 (arbitrary units). Find the Rosseland and Planck means. (c) Which mean controls the energy flux in a stellar interior, and why?

> **Solution.** (a) $F = 2\pi \int_{-1}^{1} (B + \mu B') \mu \, d\mu = 2\pi B' \cdot \tfrac{2}{3} = \tfrac{4\pi}{3} B'$. (b) $1/\kappa_R = 0.5/1 + 0.3/10 + 0.2/1000 = 0.5302 \Rightarrow \kappa_R = 1.89$. $\kappa_P = 0.5 + 3 + 200 = 203.5$. (c) The Rosseland mean, because the flux is $\propto \sum (1/\kappa_\nu)$: radiation leaks through the most transparent band, so $\kappa_R \approx 2\times$ the window opacity, while the Planck mean is dominated by the most opaque band.

Part C — Formation of the RTE, polarization, history (4 marks each)

C1. (Lecture RT3) Show how the term $\frac{1-\mu^2}{r}\frac{\partial I}{\partial \mu}$ arises in the spherical RTE, and state when it may be neglected.

> **Solution.** Along a straight ray with impact parameter p: $r^2 = p^2 + z^2$, $\mu = z/r$, so $dr/dz = \mu$ and $d\mu/dz = (1-\mu^2)/r$. Then $dI/dz = \mu\, \partial_r I + \frac{1-\mu^2}{r} \partial_\mu I$. The term describes the change of a ray's angle to the local vertical. It is negligible when the atmosphere is thin compared with r (about $10H \ll R$), e.g. the Sun.

C2. (Lecture RT6) Schwarzschild (1906) used two streams at $\mu = \pm \tfrac{1}{2}$ and found $T^4 = \tfrac{1}{2} T_{\rm eff}^4 (1 + 2\tau)$. Derive the corresponding limb-darkening law and compare $I(0)/I(1)$ with the exact gray value 0.344.

Solution. $S \propto 1 + 2\tau$ is linear, so Eddington–Barbier is exact: $I(0, \mu) \propto 1 + 2\mu$ and $I(\mu)/I(1) = (1 + 2\mu)/3$. At the limb, $I(0)/I(1) = 1/3 = 0.333$, only 3 % below the exact 0.344 (Eddington's approximation gives 0.400).

C3. (Lecture RT5) Light is measured through filters: $I_1 = 0.55$, $I_2 = 0.45$, $I_{1'} = 0.50$, $I_{2'} = 0.50$, $I_+ = 0.52$, $I_- = 0.48$ (units of the total). Find p, p_L and p_V. Which physical process could produce circular polarization in a stellar line?

Solution. $I = 1$, $Q = 0.10$, $U = 0$, $V = 0.04$. Then $p = \sqrt{0.01 + 0.0016} = 0.108$, $p_L = 0.10$, $p_V = 0.04$. Circular polarization in a spectral line comes from the Zeeman effect with a magnetic field component along the line of sight ($V \propto B \cos\gamma \, \partial I/\partial\lambda$ in the weak-field limit).

Part E — Essay (10 marks)

E1. Starting from photon conservation, explain how the radiative transfer equation is derived, which assumptions lead to its plane-parallel static form, and how the main historical approximations (two-stream, Eddington, diffusion) and exact methods (Hopf, Chandrasekhar) solve it.

Model answer outline. (1) Photon distribution f, $I_\nu = (h^4\nu^3/c^2) f$; Boltzmann equation; gains and losses (absorption with stimulated emission, emission, scattering out and in with redistribution function). (2) Assumptions: static, straight rays, plane-parallel, so $\mu \, dI/d\tau = I - S$. (3) Formal solution and Schwarzschild–Milne equations; scattering makes the problem an integral equation. (4) Approximations: two-stream (Schuster, Schwarzschild 1906: $T^4 = \frac{1}{2} T_{\rm eff}^4 (1 + 2\tau)$), Eddington $K = J/3$, $J(0) = 2H$, diffusion and Rosseland mean deep inside. (5) Exact: Hopf $q(\tau)$, Chandrasekhar's discrete ordinates and H-functions (limb darkening 0.344 at the limb). (6) Numerical era: Feautrier, Λ-iteration and its failure, ALI, short characteristics, Monte Carlo.

Formula Sheet — Course A: Radiative Transfer

Radiation field (Lecture RT1)

Specific intensity	$dE = I_\nu \cos\theta \, dA \, d\Omega \, d\nu \, dt$; I_ν constant along rays in vacuum; I_ν/ν^3 invariant
Mean intensity	$J_\nu = \frac{1}{4\pi}\oint I_\nu d\Omega = \frac{1}{2}\int_{-1}^{1} I_\nu d\mu$
Flux	$F_\nu = \oint I_\nu \cos\theta \, d\Omega = 2\pi \int_{-1}^{1} I_\nu \mu \, d\mu$; $H_\nu = F_\nu/4\pi$
K-integral, pressure	$K_\nu = \frac{1}{2}\int_{-1}^{1} I_\nu \mu^2 d\mu$; $P_\nu = 4\pi K_\nu/c$
Energy density	$u_\nu = 4\pi J_\nu/c$
Isotropic field	$F = 0, K = J/3, P = u/3$
Uniform hemisphere	$F = \pi I$
Sphere seen from d	$F = \pi I (R/d)^2$
$\nu \leftrightarrow \lambda$	$I_\lambda = (c/\lambda^2) I_\nu$

Black body (Lecture RT2)

$$B_\nu = \frac{2h\nu^3}{c^2}\frac{1}{e^{h\nu/kT}-1}, \quad B_\lambda = \frac{2hc^2}{\lambda^5}\frac{1}{e^{hc/\lambda kT}-1}$$

- RJ: $B_\nu \approx 2\nu^2 kT/c^2$ · Wien: $B_\nu \approx \frac{2h\nu^3}{c^2} e^{-h\nu/kT}$ · $h\nu_{\max} = 2.821kT \cdot \lambda_{\max} T = 2.898 \times 10^{-3}$ m K
- $\int B_\nu d\nu = \sigma T^4/\pi \cdot F = \sigma T^4 \cdot u = aT^4, a = 4\sigma/c \cdot dB/dT = 4\sigma T^3/\pi \cdot n_\gamma = 2.03 \times 10^7 T^3$ m^{-3}
- T_b: $I_\nu = B_\nu(T_b)$, RJ $T_b = c^2 I_\nu/2k\nu^2 \cdot L = 4\pi R^2 \sigma T_{\text{eff}}^4$

Coefficients (Lecture RT2)

- $\alpha_\nu = n\sigma_\nu = \rho\kappa_\nu \cdot \ell = 1/\alpha_\nu \cdot \tau_\nu = \int \alpha_\nu ds \cdot S_\nu = j_\nu/\alpha_\nu \cdot \kappa_{\text{es}} = 0.02(1+X)$ m^2 kg^{-1}
- Einstein: $g_1 B_{12} = g_2 B_{21}$, $A_{21} = \frac{2h\nu^3}{c^2} B_{21}$
- $j_\nu = \frac{h\nu}{4\pi} n_2 A_{21} \phi$, $\alpha_\nu = \frac{h\nu}{4\pi}\phi(n_1 B_{12} - n_2 B_{21})$, $S = \frac{2h\nu^3/c^2}{g_2 n_1/g_1 n_2 - 1}$
- Scattering + absorption: $S = \varepsilon B + (1-\varepsilon)J$ · Random walk: $R = \sqrt{N}\ell$, $N \approx \max(\tau, \tau^2)$

Transfer equation (Lectures RT3, RT4)

- $\frac{dI_\nu}{d\tau_\nu} = S_\nu - I_\nu$ (τ along the ray) · $I = I_0 e^{-\tau} + \int_0^\tau S(t) e^{-(\tau - t)} dt$ · slab: $I = I_0 e^{-\tau} + S(1 - e^{-\tau})$
- Plane-parallel (τ inward): $\mu \frac{dI}{d\tau} = I - S \cdot I(0, \mu) = \int_0^\infty S \, e^{-t/\mu} dt/\mu$

- $J = \frac{1}{2}\int_0^\infty S\,E_1|t-\tau|dt \cdot F = 2\pi\left[\int_\tau^\infty SE_2(t-\tau)dt - \int_0^\tau SE_2(\tau-t)dt\right] \cdot K = \frac{1}{2}\int SE_3|t-\tau|dt$
- $E_n(x) = \int_1^\infty e^{-xt}t^{-n}dt$, $E_n' = -E_{n-1}$, $E_n(0) = 1/(n-1)$, $\int_0^\infty E_n = 1/n$, $\int_0^\infty tE_n = 1/(n+1)$
- **Eddington–Barbier:** $I(0,\mu) \approx S(\tau=\mu)$, $F(0) \approx \pi S(\tau=2/3)$ (exact for linear S)

Moments, diffusion, scattering (Lecture RT4)

- $\dfrac{dH}{d\tau} = J - S \cdot \dfrac{dK}{d\tau} = H$ · RE: $\int \kappa_\nu(J_\nu - B_\nu)d\nu = 0$, $F = \sigma T_{\text{eff}}^4$ = const
- Eddington: $K = J/3$, $J(0) = 2H(0) \Rightarrow \frac{1}{3}J'' = J - S$
- Diffusion: $F_\nu = \frac{4\pi}{3}\frac{dB_\nu}{d\tau_\nu} \cdot \dfrac{1}{\kappa_R} = \dfrac{\int \kappa_\nu^{-1}\partial_T B_\nu d\nu}{\int \partial_T B_\nu d\nu} \cdot F = -\dfrac{16\sigma T^3}{3\kappa_R\rho}\dfrac{dT}{dz}$
- Isothermal scattering: $S(0) = \sqrt{\varepsilon}\,B$ (exact) · Eddington $J = B\left[1 - \frac{e^{-\sqrt{3\varepsilon}\tau}}{1+2\sqrt{\varepsilon/3}}\right]$ · $\tau_{\text{th}} \sim 1/\sqrt{\varepsilon}$

Forming the RTE (Lecture RT3)

- $I_\nu = (h^4\nu^3/c^2)f$ · Boltzmann: $\partial_t f + c\hat{\mathbf{n}}\cdot\nabla f = (\partial_t f)_{\text{coll}}$
- General: $\frac{1}{c}\partial_t I_\nu + \hat{\mathbf{n}}\cdot\nabla I_\nu = -(\kappa_\nu+\sigma_\nu)\rho I_\nu + j_\nu + \frac{\sigma_\nu\rho}{4\pi}\iint R I_{\nu'}d\Omega' d\nu'$
- Spherical: $\mu\partial_r I + \frac{1-\mu^2}{r}\partial_\mu I = \chi(S-I)$ · invariants I_ν/ν^3, $\nu\chi_\nu$, j_ν/ν^2
- Energy: $\partial_t u + \nabla\cdot\mathbf{F} = 4\pi\int(j_\nu - \kappa_\nu\rho J_\nu)d\nu$ · force: $\mathbf{f}_{\text{rad}} = \frac{1}{c}\int \chi_\nu \mathbf{F}_\nu d\nu$

Historical approximations (Lecture RT6)

- Schwarzschild ($\mu = \pm\frac{1}{2}$): $T^4 = \frac{1}{2}T_{\text{eff}}^4(1+2\tau)$, $I(\mu)/I(1) = (1+2\mu)/3$
- Two-stream ($\mu = \pm 1/\sqrt{3}$): $T^4 = \frac{3}{4}T_{\text{eff}}^4(\tau + 1/\sqrt{3})$ · Eddington: $\frac{3}{4}T_{\text{eff}}^4(\tau + \frac{2}{3})$ · exact: $\frac{3}{4}T_{\text{eff}}^4(\tau + q)$

Polarization (Lecture RT5)

- $I = I_1 + I_2$, $Q = I_1 - I_2$, $U = I_{1'} - I_{2'}$, $V = I_+ - I_-$ · $p = \sqrt{Q^2+U^2+V^2}/I$ · $\chi = \frac{1}{2}\arctan(U/Q)$
- $d\mathbf{I}/ds = -\mathsf{K}\mathbf{I} + \boldsymbol{\epsilon}$ (Unno–Rachkovsky with Zeeman profiles)
- Thomson/Rayleigh: $p = \sin^2\Theta/(1+\cos^2\Theta)$ · electron-scattering atmosphere: $p(\mu=0) = 11.7\,\%$
- Zeeman: $\Delta\lambda_B = 4.67\times10^{-13} g\lambda^2 B$ (λ, Δλ in Å; B in G) · weak field: $V \approx -\Delta\lambda_B\cos\gamma\,\partial I/\partial\lambda$